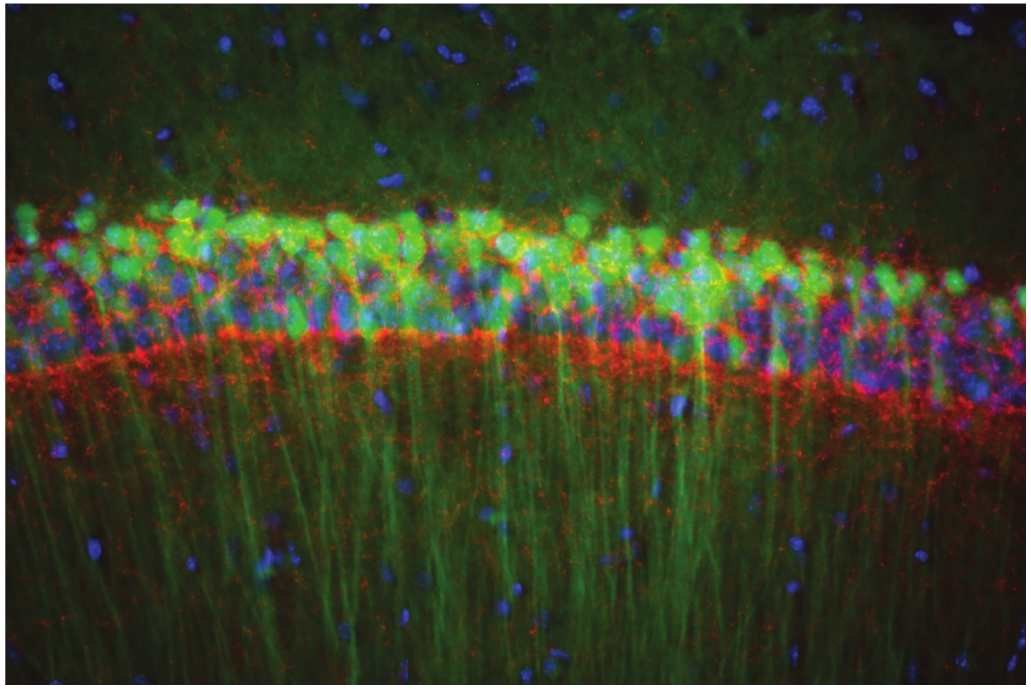


Carlson and Birkett (2022), Chapter 19 opener, p.604. Image: BSIP/Newscom.

BEHAVIORAL NEUROSCIENCE 392

Substance Abuse

From Reward to Compulsion: Drugs, Learning, and Recovery

Fall 2026

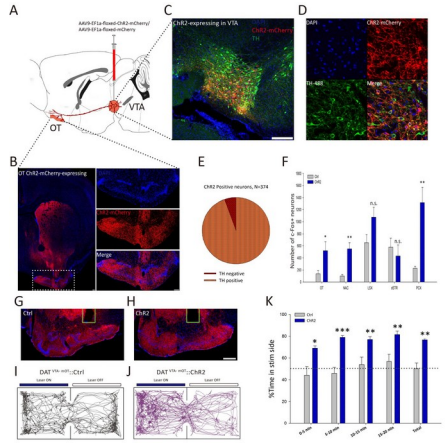
Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

Substance use disorder involves impaired control

A substance use disorder is a persistent pattern of use that causes clinically significant impairment or distress. Features can include impaired control, strong motivation to obtain the substance, and continued use despite harmful consequences. The diagnosis concerns behavior and functioning, not simply whether someone has ever used a drug.

Physical dependence means the body has adapted so that stopping or reducing exposure produces withdrawal. Tolerance means a given amount produces a smaller effect after repeated exposure. Either can occur during appropriate medical treatment and does not by itself establish compulsive use.

Researchers isolate reinforcement and learning using tasks with defined consequences. The mouse shown here explores an apparatus before reward-related stimulation is introduced. This baseline measures where it normally moves, allowing later preference to be distinguished from a pre-existing attraction to one location.



Zhang et al. (2017), Fig. 3.



Exploration before place-preference training. Zhang et al. (2017).

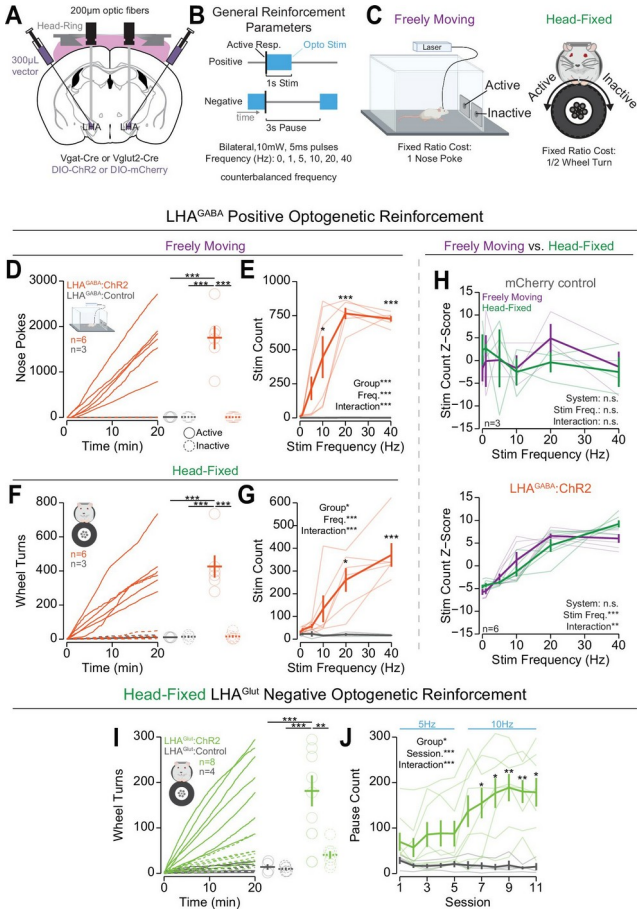
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Reinforcement changes the probability of an action

Positive reinforcement strengthens an action because it produces a desirable consequence. Negative reinforcement strengthens an action because it removes an unpleasant condition. Both processes increase future responding, even though one adds an outcome and the other removes one.

Punishment instead reduces an action through its consequences. The word negative in negative reinforcement therefore does not mean punishment or harm. For example, relief of withdrawal can strengthen drug taking because the action temporarily removes an aversive bodily state.

Reinforcement is defined by the observed change in behavior rather than the researcher's intention. An offered reward may fail to strengthen responding, while an apparently unpleasant event can have different effects in different circumstances. The action and its contingent consequence must be measured together.



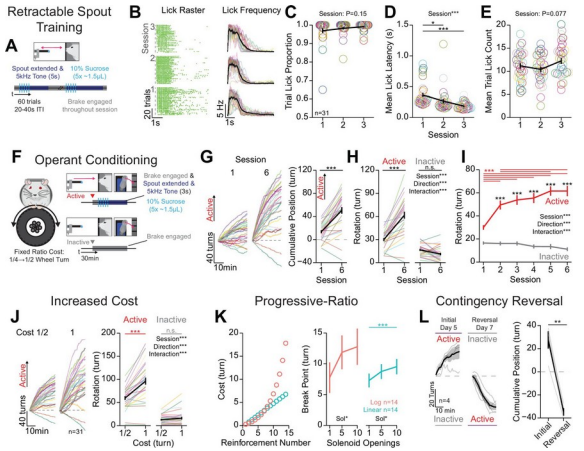
Gordon-Fennell et al. (2023), Fig. 3.

An operant action earns a defined reward

OHRBETS is an experimental system for studying reward and aversion in head-fixed mice. The animal can turn a small lever while receiving controlled outcomes. Head fixation stabilizes recordings, while the lever preserves an action whose consequence can be changed by the experimenter.

The original recording follows lever turns that earn sucrose under fixed-ratio schedules. A fixed ratio specifies how many responses are required for an outcome. Changing that requirement alters the action cost while leaving the basic reward and response available to the animal.

The study combined such tasks with measurements of dopamine-related activity and consumption. This makes the action, outcome delivery, and neural signal separately observable. The selected demonstration concerns food reinforcement, providing a defined learning process rather than a recording of compulsive drug taking.



Gordon-Fennell et al. (2023), Fig. 2.



Lever responding for sucrose. Gordon-Fennell et al. (2023).

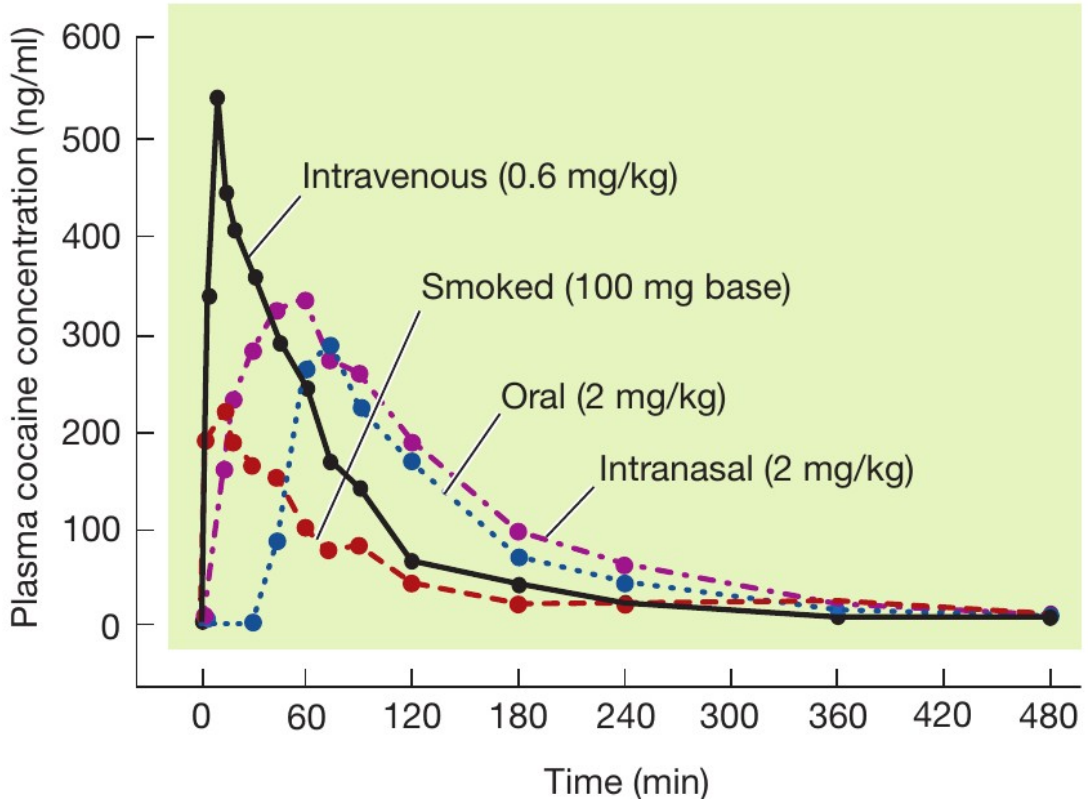
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The speed of drug delivery changes reinforcement

A psychoactive drug changes nervous-system function after reaching relevant targets. Its effects depend on absorption, distribution, metabolism, and removal, as well as receptor action. The same molecule can therefore have different time courses when it enters the body by different routes.

Rapid delivery can produce a steep rise in brain concentration and make an action's consequence closely follow that action. Close timing helps reinforcement and cue learning. Longer-lasting exposure produces a different pattern of receptor occupancy even when the total amount reaching the body is comparable.

Delivery kinetics do not determine all risk independently of the person or setting. Prior learning, availability, stress, and biological sensitivity also shape continued use. The mechanistic distinction is between how quickly a drug reaches its target and what it does after arriving there.



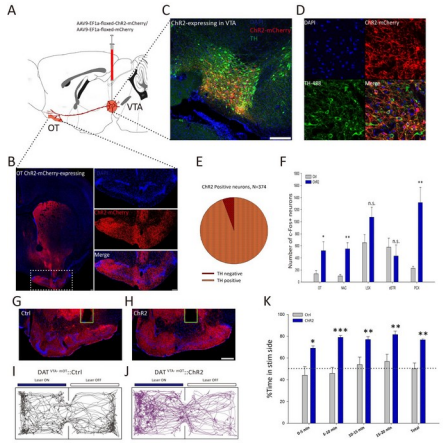
Carlson and Birkett (2022), Fig. 19.1, p.608.

A learned place preference persists without stimulation

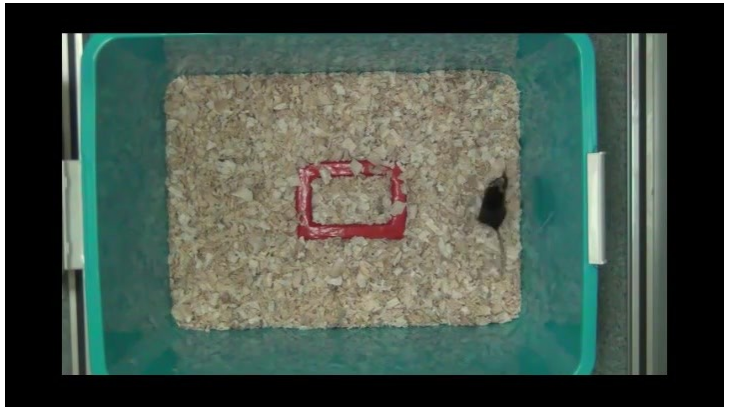
Researchers stimulated dopamine-containing axons from the ventral tegmental area to the medial olfactory tubercle in mice. During training, stimulation occurred in a designated part of the apparatus. The medial olfactory tubercle is a reward-related forebrain region that also receives odor information.

The selected post-training recording tests location preference without ongoing light delivery. Mice returned preferentially to the region previously associated with stimulation. The response therefore depended on the learned value of the location rather than light forcing movement at that instant.

The study compared this test with pre-training exploration in the same apparatus. Measuring both stages separated acquired preference from ordinary spatial bias. The specific projection terminates in medial olfactory tubercle, so this experiment should not be described as stimulation of nucleus accumbens terminals.



Zhang et al. (2017), Fig. 3.



Place-preference test without stimulation. Zhang et al. (2017).

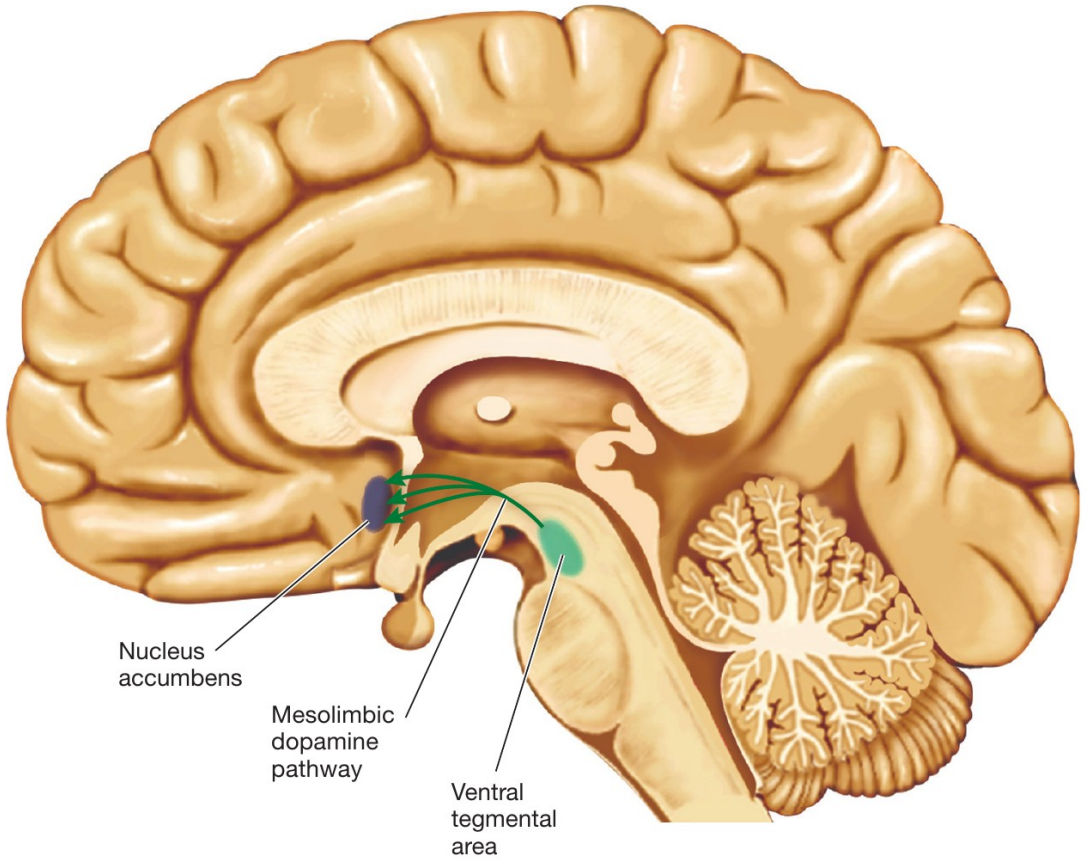
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Mesolimbic dopamine helps reinforce learned actions

The ventral tegmental area, or VTA, contains dopamine neurons that project to forebrain regions including the nucleus accumbens. The accumbens is part of the ventral striatum and helps translate motivational information into action. Dopamine changes how these circuits respond to inputs and learn from outcomes.

Many reinforcing drugs increase dopamine signaling through different initial targets. Some act on transporters, while others alter neurons that control dopamine-cell firing. A shared downstream change therefore does not mean that every drug binds the same receptor or produces the same physiological effects.

Dopamine is not a direct meter of pleasure. Its activity varies with predictions, salience, movement, and other task conditions. To interpret a dopamine signal, researchers must identify its location, the event that changed it, and the behavior being measured at that time.



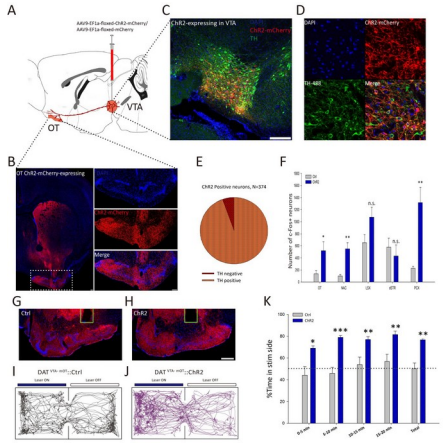
Carlson and Birkett (2022), Fig. 19.2, p.609.

Stimulation can establish a location's value

In the first training recording, entering the designated apparatus region triggers stimulation of VTA axons in medial olfactory tubercle. The consequence is spatially contingent: the animal's position controls when the light is delivered. This permits the location to acquire value through repeated experience.

The investigators later tested the location without stimulation and found a learned preference. Training and testing therefore address different stages. The training recording shows the contingency being experienced, while the later test measures whether the animal retains an attraction to the previously paired region.

Stimulating an identified dopamine projection bypasses the initial pharmacological targets through which drugs normally act. It tests the contribution of that pathway to reinforcement-related learning. The preparation establishes a causal role for the projection without requiring administration of a reinforcing drug in this demonstration.



Zhang et al. (2017), Fig. 3.



First location-contingent training session. Zhang et al. (2017).

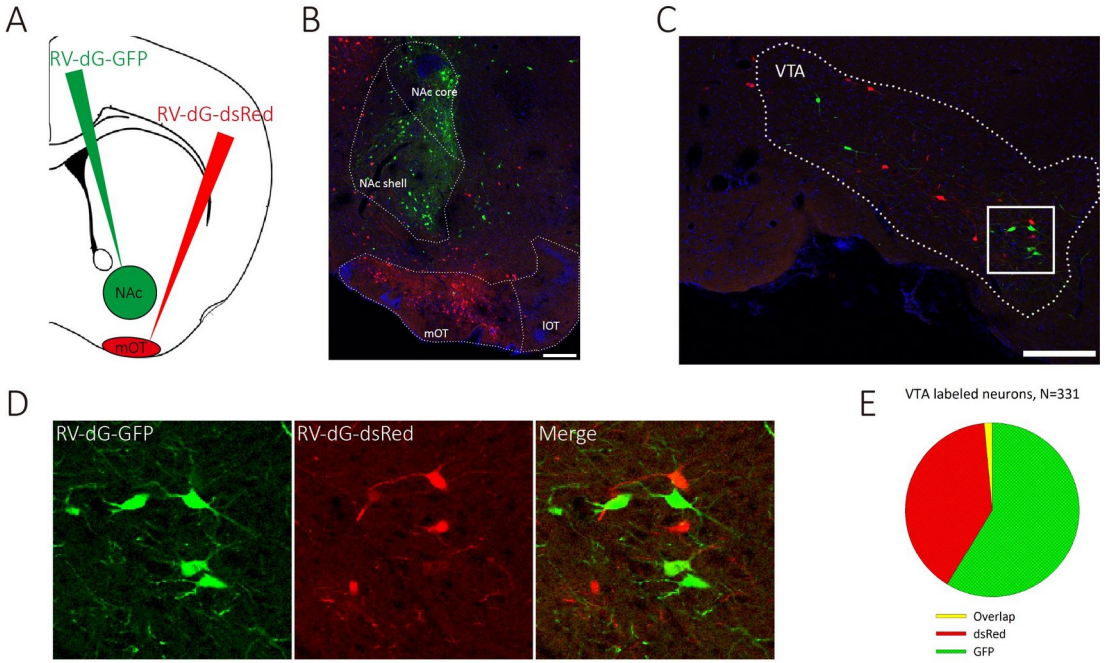
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Striatal dopamine receptors regulate different cells

D1-like and D2-like receptors are major families through which dopamine changes cellular signaling. They influence enzymes that regulate cyclic AMP, an intracellular messenger. Their effects depend on the cell and its other inputs rather than directly encoding one universal good or bad outcome.

The ventral striatum contributes to motivation and reinforcement, while dorsal-striatal systems participate in action selection and well-practiced behavior. Repeated drug experience can alter activity and plasticity across these systems. Compulsive responding is therefore not explained by one brief dopamine increase alone.

The prefrontal cortex contributes to evaluating consequences and controlling actions according to goals. Its development and current physiological state influence these functions. Drug-related behavior reflects interactions among motivational signals, learned actions, and control systems rather than a complete loss of all deliberate choice.



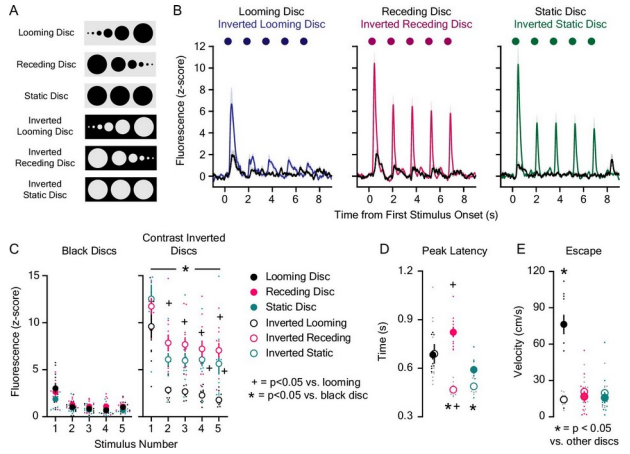
Zhang et al. (2017), Fig. 6.

A looming stimulus can change accumbens dopamine

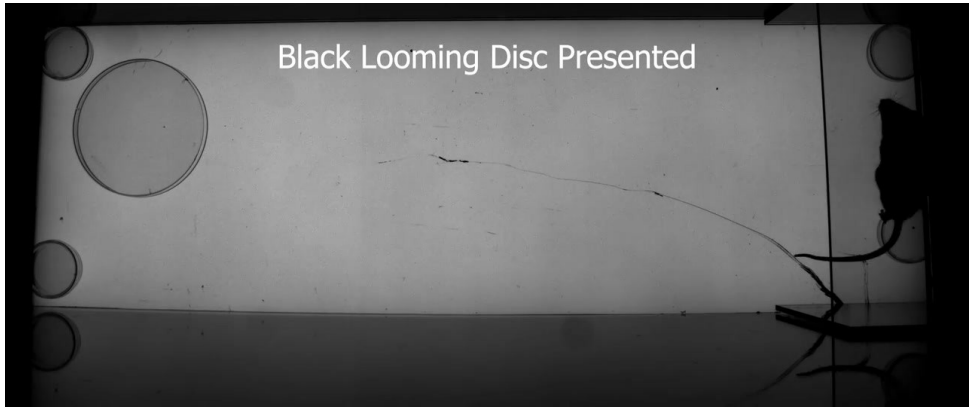
A mouse study measured dopamine signals in the lateral nucleus accumbens during visual events. The researchers used dLight, a fluorescent sensor whose brightness changes when dopamine binds. This measured local dopamine-related fluorescence while the mouse encountered different patterns of illumination.

The original looming-stimulus recording shows the animal's movement during an expanding visual event. Such a stimulus can resemble an approaching overhead object and evoke defense. The investigators compared its sensory features with other visual presentations rather than treating any dopamine response as evidence of pleasure.

The dopamine signal tracked the rate and magnitude of illumination change more closely than the inferred threat level. A response to the visual event therefore required a sensory explanation. This finding separates a measured transmitter signal from the emotional label assigned to the event by an observer.



Gonzalez et al. (2023), Fig. 1.



Mouse behavior during a looming stimulus. Gonzalez et al. (2023).

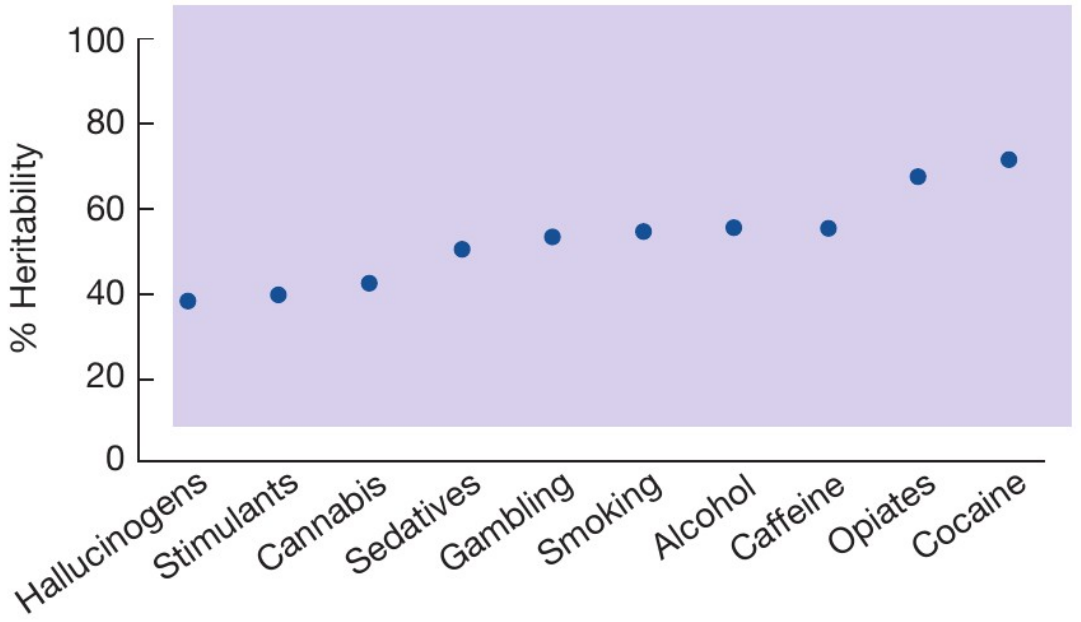
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Adolescence and environment influence vulnerability

Adolescence includes continuing development of circuits involved in motivation, learning, and behavioral control. Experiences during this period can alter later responses to rewards and cues. Increased vulnerability does not mean that every adolescent exposure inevitably produces a substance use disorder.

Genetic variation can influence drug metabolism, receptor function, or sensitivity to effects. Environmental conditions influence availability, stress, social learning, and alternative rewards. Risk develops through their interaction rather than one gene determining a fixed outcome independently of experience.

Repeated exposure also changes the nervous system through learning and cellular adaptation. These acquired changes must be separated from differences present before use began. A study that measures people only after a disorder develops cannot by itself establish which biological differences preceded the repeated exposure.



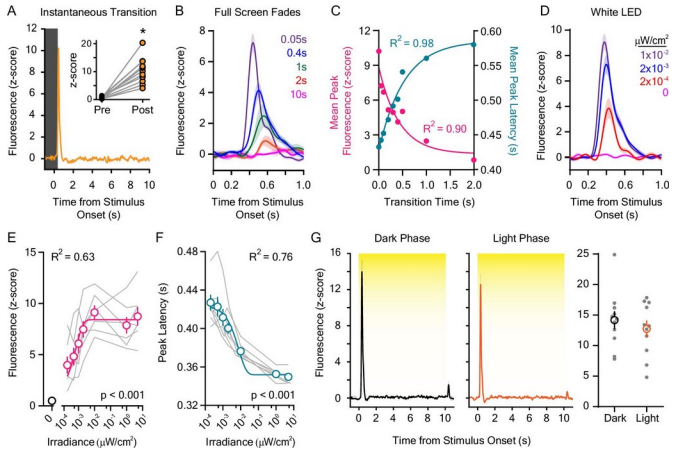
Carlson and Birkett (2022), Fig. 19.10, p.616.

Visual timing changes the measured dopamine response

The accumbens study systematically varied the visual stimulus rather than presenting only one threatening image. The selected original stimulus movie shows the patterns used in the experiments. It is the publisher's task presentation, separate from the accompanying recordings of freely moving animals.

Changing how quickly illumination rose or fell changed the dopamine-related response. Wavelength also affected the response, consistent with sensory processing contributing to the signal. These comparisons tested properties of the input that would remain hidden if all stimuli were classified only as threatening or neutral.

The researchers also examined pathway-related controls to establish that the fluorescent response was not simply an arbitrary movement trace. The central finding remained stimulus-specific: local dopamine tracked defined changes in illumination. This is a measured sensory response, not proof that dopamine has no role in reinforcement elsewhere.



Gonzalez et al. (2023), Fig. 2.



Original visual-stimulus sequence. Gonzalez et al. (2023).

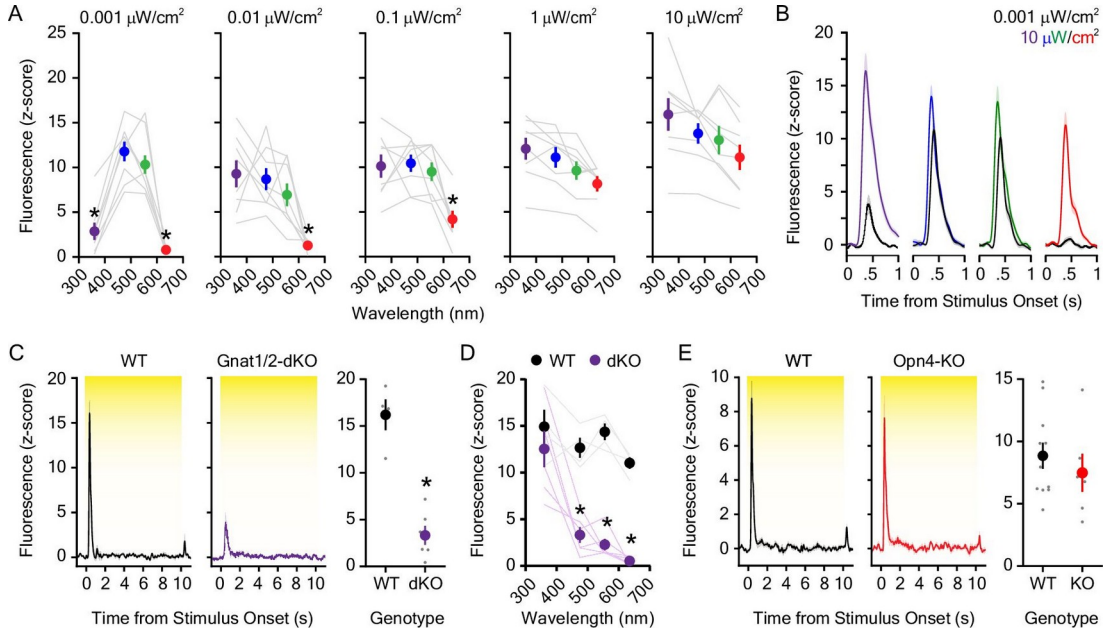
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A transmitter response need not report hedonic value

Hedonic value concerns how pleasant or unpleasant an experience is. Incentive value concerns how strongly it attracts behavior, and a sensory response concerns detection of an event's features. These quantities can vary together in some tasks without being interchangeable measurements.

The lateral-accumbens experiment dissociated visual features from the assumed threatening significance of the stimulus. Its dopamine signal changed with illumination dynamics even when a pleasure interpretation would be implausible. The anatomical location and experimental event specified what was measured.

This does not require abandoning dopamine's established roles in reinforcement. Different projections and local circuits can carry different mixtures of information. A drug-induced dopamine increase must therefore be interpreted within the relevant circuit and learned behavior, rather than equating fluorescence amplitude with enjoyment.



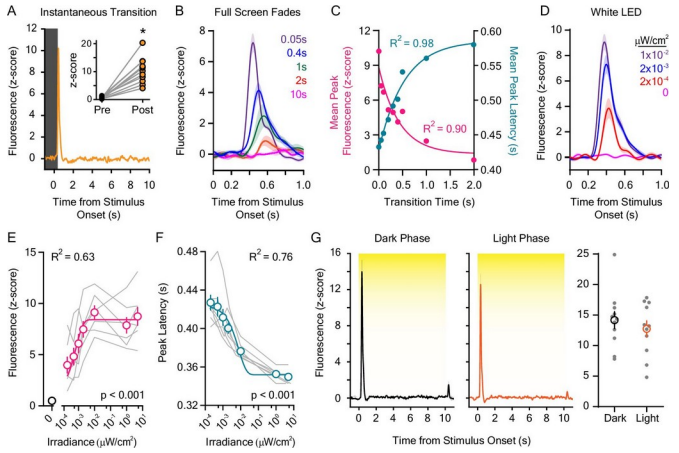
Gonzalez et al. (2023), Fig. 4.

A spotlight provides a different visual challenge

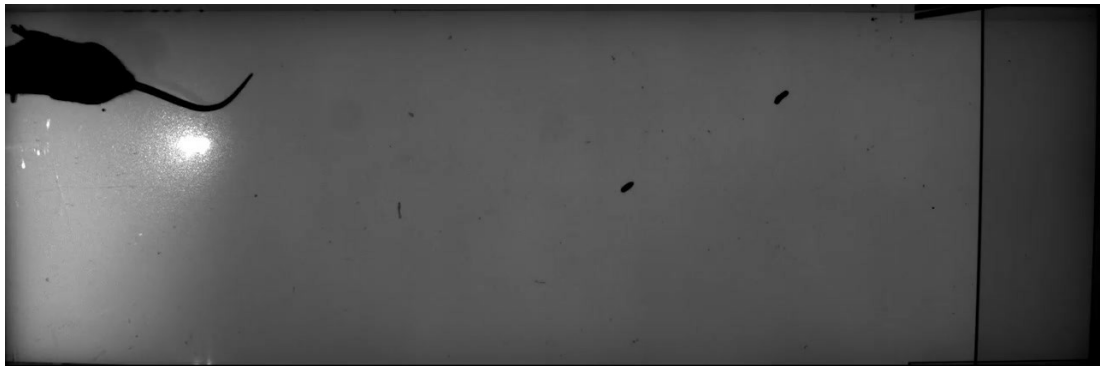
The selected spotlight recording follows a mouse during a localized illumination change. It provides a real behavioral comparison with the looming stimulus. The two visual events differ in their spatial and temporal structure, allowing sensory properties to be assessed alongside the animal's movement.

The study found that dopamine in lateral accumbens followed illumination dynamics across these comparisons. The response could not be explained simply by assigning a larger threat value to every stimulus with a larger signal. The measured relationship concerned how the incoming light changed.

The animal's locomotion and the transmitter measurement answer different questions. One describes the action that occurred during the visual event; the other follows a chemical signal in a defined region. Keeping these measurements separate prevents movement or arousal from being silently renamed as pleasure.



Gonzalez et al. (2023), Fig. 2.



Mouse behavior during spotlight illumination. Gonzalez et al. (2023).

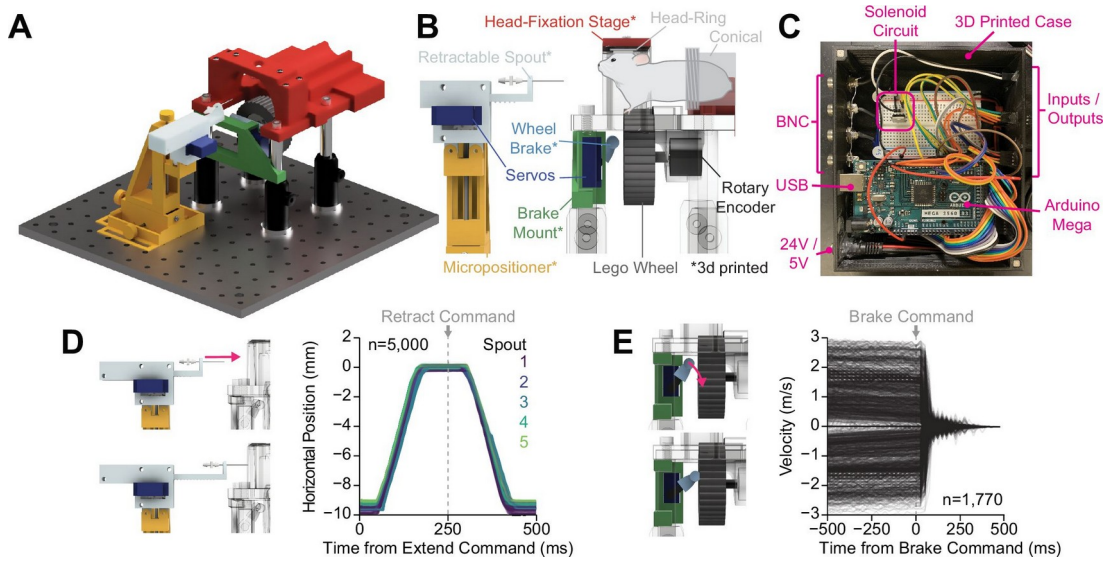
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One apparatus separates distinct reward measures

The OHRBETS researchers designed tasks that separated action, delivery, and consumption. A mouse could perform the lever response before receiving the outcome, and licking provided another measure after delivery. These events do not necessarily occur at the same moment or reflect the same motivational process.

The system also allowed outcomes to vary in value or aversiveness while recording behavior under stable imaging conditions. Comparing these phases helped identify whether a neural signal accompanied seeking, receiving, or consuming. A response during consumption cannot automatically be assigned to anticipation of the outcome.

Head fixation changes the available movements, but the animal still controls a defined action-outcome contingency. The experimental claim concerns that contingency and the recorded signals. It does not require treating every lever turn as equivalent to unrestricted seeking of a drug in a natural environment.



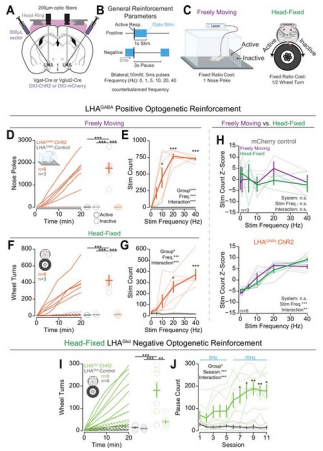
Gordon-Fennell et al. (2023), Fig. 1.

An action can gain stimulation or terminate it

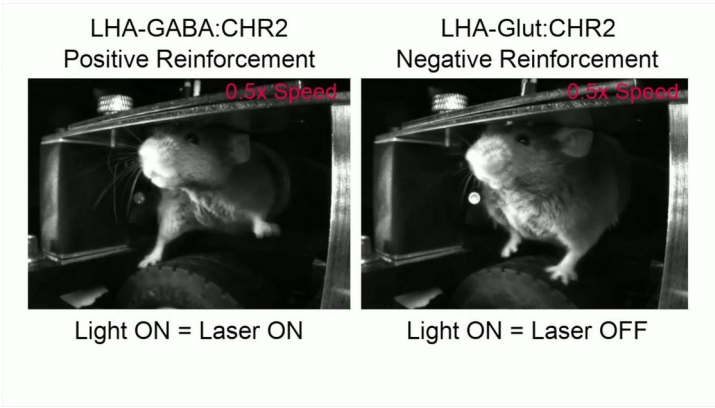
The original paired OHRBETS demonstration contains two reinforcement arrangements. In one, responding obtains optical stimulation of lateral-hypothalamic GABA neurons. In the other, responding turns off stimulation of a lateral-hypothalamic glutamate population. The consequence of the action differs between the two panels.

The first arrangement tests positive reinforcement through obtaining an outcome. The second tests negative reinforcement through removal of an ongoing aversive condition. Responding to terminate stimulation is therefore different from receiving stimulation as a punishment for the response.

The same basic action can be strengthened by either consequence. This makes the direction of the contingency essential for interpreting the behavior. The demonstration provides a direct experimental distinction relevant to drug seeking for reward versus drug seeking to relieve an unpleasant state.



Gordon-Fennell et al. (2023), Fig. 3.



Obtaining versus terminating hypothalamic stimulation. Gordon-Fennell et al. (2023).

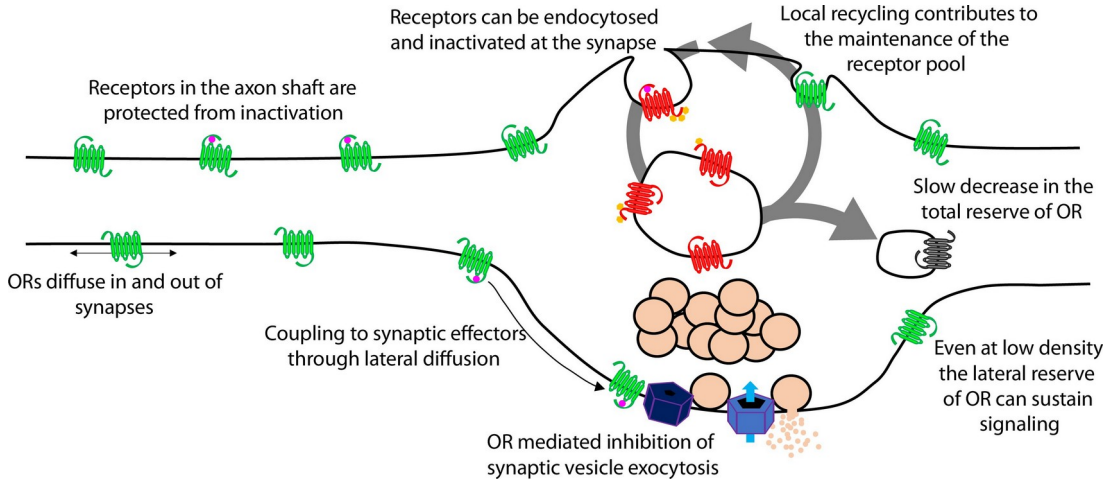
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Withdrawal can supply negative reinforcement

With repeated exposure, nervous-system adaptations can oppose a drug's acute effects. When drug concentrations fall, those adaptations can contribute to withdrawal. The symptoms depend on the drug and can include bodily discomfort, altered mood, sleep disturbance, and strong motivation to restore the previous state.

Taking the drug again may temporarily reduce these symptoms. Relief then negatively reinforces the action, adding a maintaining process beyond the initial rewarding effect. The person can continue responding even when the pleasurable effect has diminished or use has become harmful.

Withdrawal and craving are related but distinct. Craving can also be triggered by cues, stress, or memories after acute withdrawal has subsided. A treatment that controls withdrawal therefore addresses an important maintaining process without necessarily eliminating every learned trigger for future use.



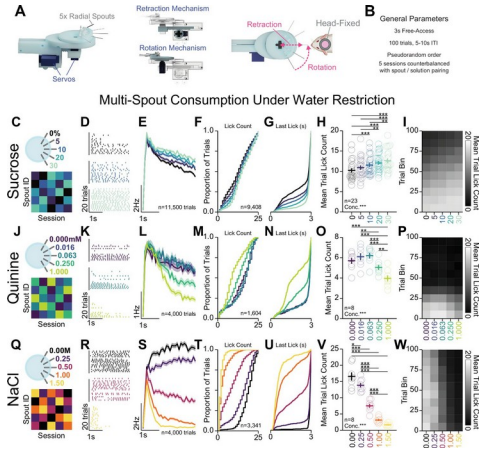
Jullié et al. (2022), Fig. 7.

Consumption varies with the available outcome

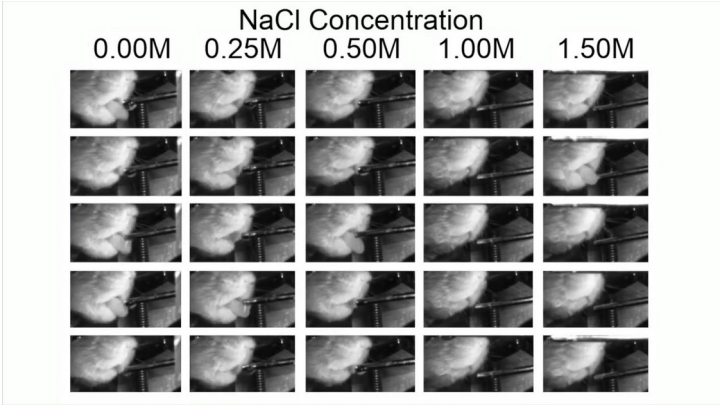
The OHRBETS multi-spout task presented solutions differing in salt concentration. The selected original recording follows licking across the available conditions. Consumption could therefore be measured directly rather than inferred solely from the work the animal performed to obtain access.

The investigators used these kinds of comparisons to examine how outcome properties influence behavior and neural signals. A change in licking reflects the animal's interaction with the delivered solution. It is a consummatory measure, distinct from lever responding before the outcome arrives.

The example concerns salt solutions rather than drug self-administration. Its relevance is the separation of seeking from consuming under controlled conditions. Changes in a dopamine-related signal can be aligned with these different events instead of assuming one value represents every phase of reinforcement.



Gordon-Fennell et al. (2023), Fig. 5.



Consumption across salt-solution conditions. Gordon-Fennell et al. (2023).

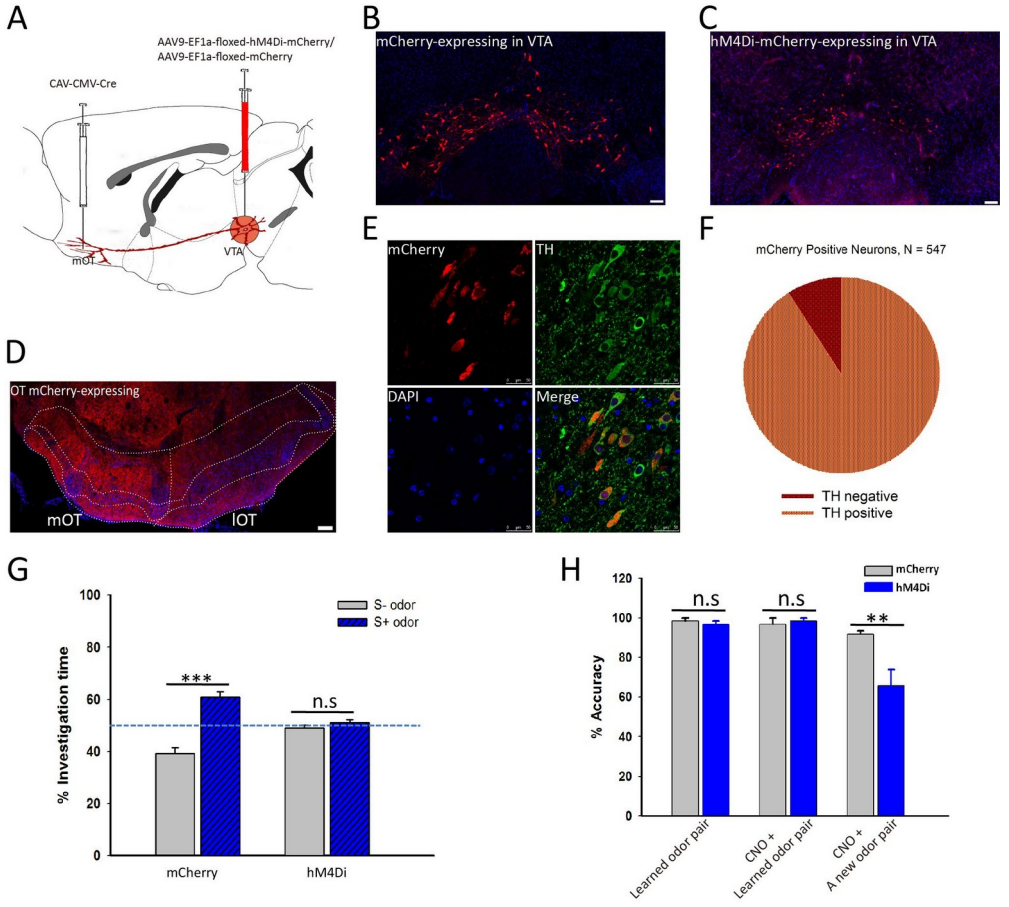
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A specific dopamine projection supports odor learning

The VTA-to-olfactory-tubercle study also paired odors with stimulation of the identified dopamine projection. An odor could therefore acquire value without being intrinsically preferred before training. The investigators tested the learned odor preference after the pairing procedure.

Blocking dopamine receptors in medial olfactory tubercle prevented the stimulation-related odor preference. This intervention acted at the terminal region rather than changing the smell itself. Dopamine signaling in that target was therefore necessary for the tested learning effect.

In a further manipulation, reducing activity in VTA neurons projecting to medial olfactory tubercle disrupted learned odor preference and odor-based discrimination learning. The projection thus contributed to both acquiring and expressing value-guided responses involving sensory information received by its forebrain target.



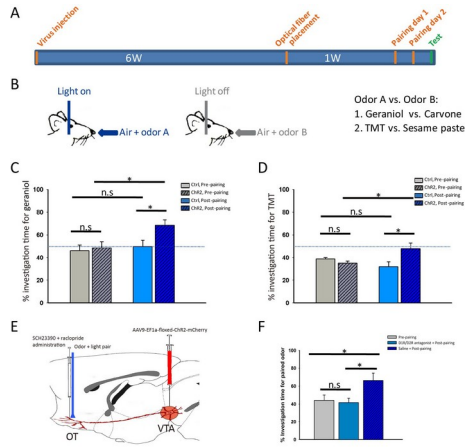
Zhang et al. (2017), Fig. 7.

An odor can retrieve an acquired preference

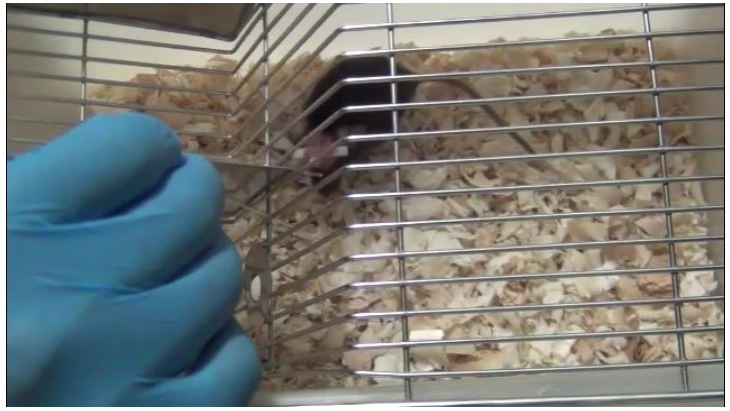
The selected original recording follows an odor-preference test after odor and dopamine-projection stimulation had been paired. The light-based training event is no longer required at every choice. The odor itself can guide where the mouse investigates because of the earlier learning.

The experiment distinguishes the sensory cue from the reinforcing event associated with it. A cue does not need to contain a drug or directly activate a receptor to influence later behavior. Its acquired value depends on the history of pairings and the neural systems that store and retrieve that history.

The study's pathway manipulations identified medial olfactory tubercle dopamine signaling as a contributor to this learning. The animal's later preference is therefore more than a response to immediate stimulation. It is an observable choice controlled by a previously neutral sensory cue.



Zhang et al. (2017), Fig. 5.



Odor preference after cue-stimulation pairing. Zhang et al. (2017).

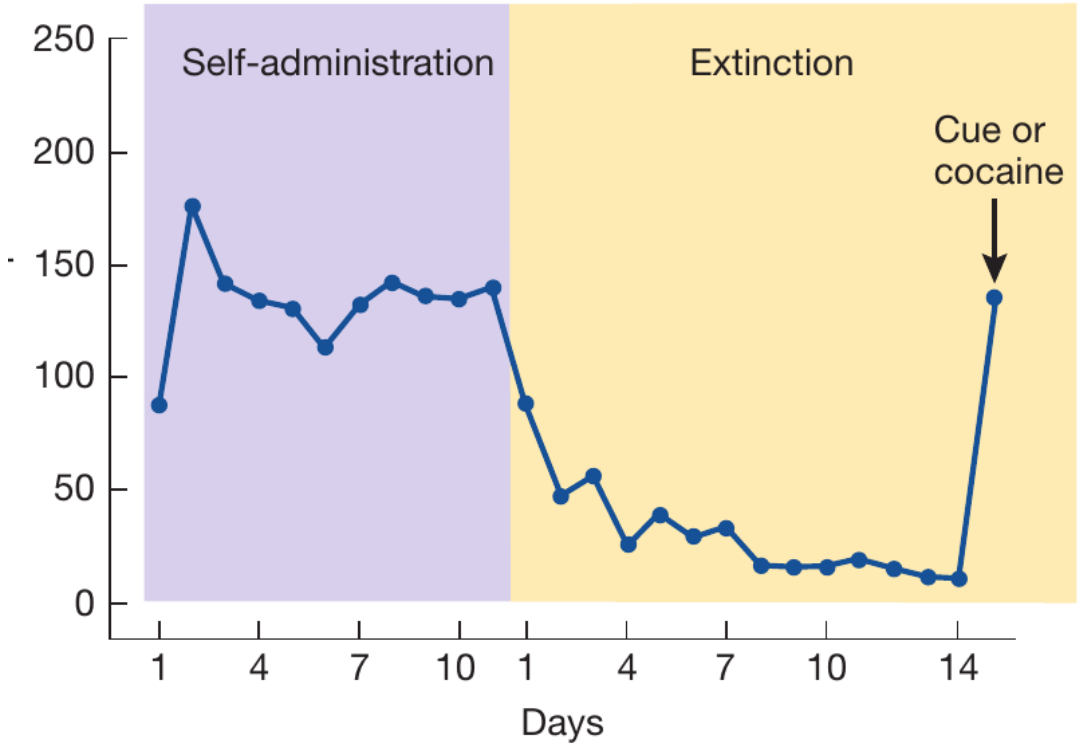
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Cues and stress can renew drug seeking

Drug-associated cues include places, objects, people, and internal sensations repeatedly paired with use. Through learning, they can attract attention and evoke expectations before the drug is present. Their influence can persist beyond the period of acute intoxication or withdrawal.

In reinstatement experiments, an animal first learns to obtain a drug and later experiences sessions without it. A cue, stressor, or drug-related event can then increase previously reduced responding. The renewed action is measured separately from whether the animal actually receives the drug during the test.

Stress can also increase the motivation to seek relief and alter regulation of learned responses. Recovery therefore includes managing cues and stress as well as drug clearance. The persistence of learned triggers helps explain why simply waiting for a substance to leave the body does not remove every source of risk.



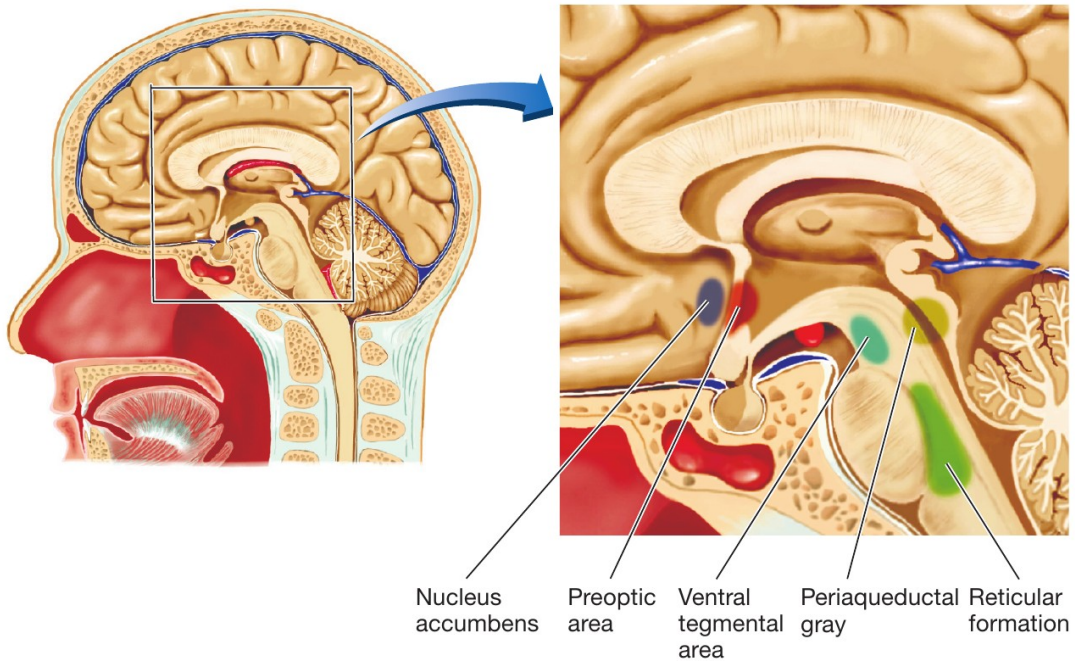
Carlson and Birkett (2022), Fig. 19.7, p.614.

Opioids act on receptors in several circuits

Opioid receptors normally respond to peptide messengers produced by the body. Mu, delta, and kappa receptors belong to a family that communicates through G proteins inside cells. Opioid drugs differ in how strongly and selectively they activate these receptor types.

In some VTA circuits, mu-receptor activation inhibits GABA-releasing cells or terminals that restrain dopamine neurons. Reducing that inhibition can increase dopamine-cell activity. This is disinhibition: inhibiting an inhibitory input allows the receiving neuron to become more active.

Opioid receptors also occur in pain pathways, brainstem circuits, and peripheral tissues. Their actions can therefore include analgesia, sedation, constipation, and respiratory suppression. Reinforcement and pain relief are different consequences of receptor activation at different cellular locations.



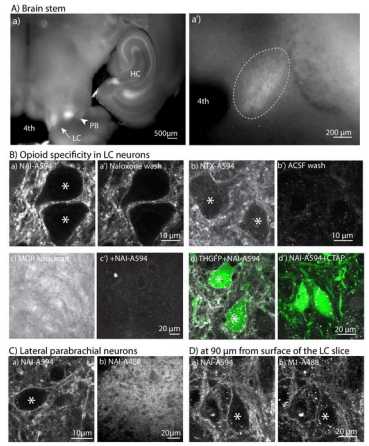
Carlson and Birkett (2022), Fig. 19.12, p.619.

Fluorescent labeling identifies mu receptors

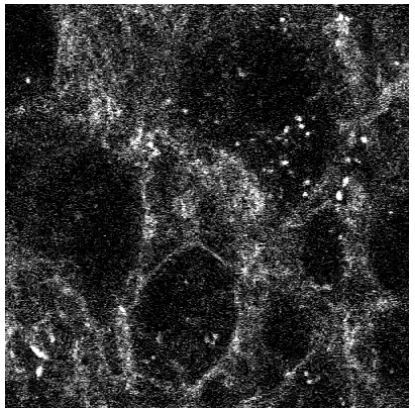
Researchers developed a probe that labels opioid receptors in living tissue without permanently occupying their drug-binding site. A receptor-binding guide brings a fluorescent tag near the receptor. After the tag is transferred, the guide can leave while the receptor remains visible.

The original depth sequence follows labeled mu receptors in the locus coeruleus of an FMOR mouse, which expresses a tagged receptor. Moving through successive optical planes reveals receptor-bearing structures within the tissue. It is a receptor-localization recording, distinct from drug-induced reinforcement or withdrawal.

The investigators tested whether labeled receptors retained function after the reaction. A probe that blocks its target would change the biology being studied. The method allowed receptor location and later physiological responses to be examined in a living preparation.



Arttamangkul et al. (2019), Fig. 3.



Mu-receptor labeling in FMOR mouse LC. Arttamangkul et al. (2019).

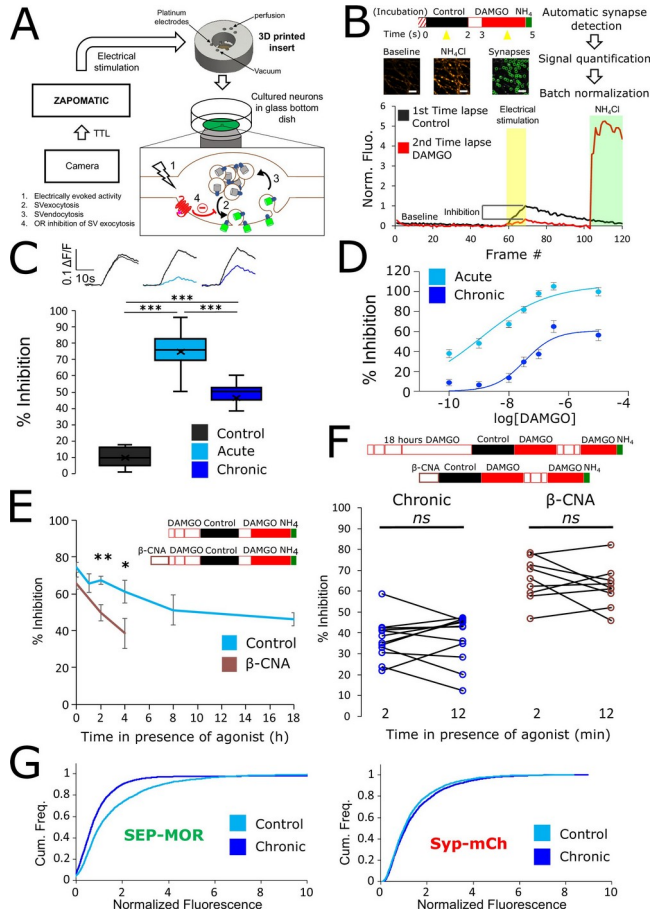
Play video

Tolerance and dependence involve adaptation

Tolerance is a reduction in a drug's effect after repeated exposure to a comparable amount. Adaptation can occur at receptors, intracellular pathways, or larger circuits. The degree of tolerance can differ among effects produced by the same drug, so one change cannot be assumed to protect against every adverse effect.

Physical dependence becomes evident when reduced exposure produces withdrawal. Cells that adapted during ongoing drug action now operate without the condition they had been counteracting. The resulting symptoms reflect altered regulation, not simply the presence of drug molecules remaining in the body.

Tolerance and dependence can coexist with a substance use disorder, but neither is its complete definition. Impaired control and continued use despite harmful consequences require separate assessment. Receptor experiments identify particular cellular adaptations rather than directly diagnosing those behavioral features.



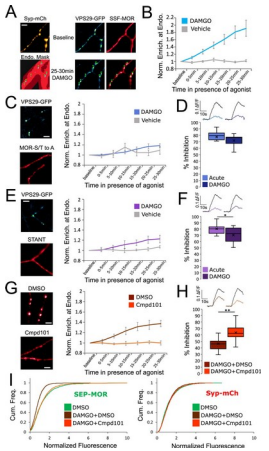
Jullié et al. (2022), Fig. 1.

Mu receptors leave the axonal surface after activation

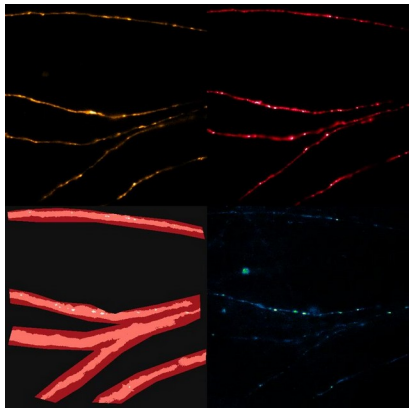
A study in cultured rat striatal neurons followed opioid receptors in axons. Fluorescent labels distinguished receptors from presynaptic structures and internal sorting compartments. Adding the mu-receptor agonist DAMGO caused receptors to move from the cell surface into the axon's interior.

This uptake is endocytosis, in which membrane and associated proteins enter internal vesicles. The original movie follows receptor redistribution alongside labeled cellular compartments. It records movement of receptor protein rather than a direct measurement of how strongly an intact animal experiences an opioid.

Over longer exposure, repeated internalization and recycling reduced the available axonal surface pool. Fewer surface receptors can reduce the response to later stimulation. The study therefore identified a trafficking mechanism that contributes to tolerance at presynaptic opioid receptors.



Jullié et al. (2022), Fig. 2.



Mu-receptor trafficking after DAMGO. Jullié et al. (2022).

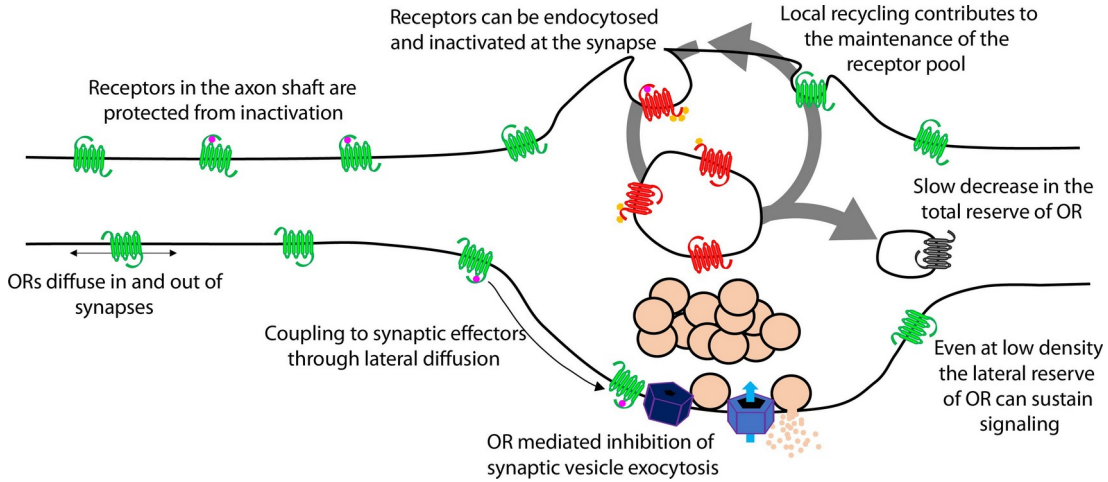
Play video

Opioid G-protein signaling reduces transmitter release

When an opioid receptor activates an inhibitory G protein, its components influence several cellular targets. They can reduce cyclic AMP production and regulate ion channels. The resulting effect depends on whether the receptor is located on a cell body, dendrite, or presynaptic terminal.

At a presynaptic terminal, reducing calcium-channel activity can decrease transmitter release. Calcium entry normally helps trigger fusion of transmitter-filled vesicles with the membrane. Less entry can therefore weaken the chemical signal sent to the next cell.

On some neurons, opioid signaling also increases potassium conductance and reduces excitability. These actions explain how receptor activation can suppress cellular output. The behavior produced by that suppression depends on what the affected cell normally does within its circuit.



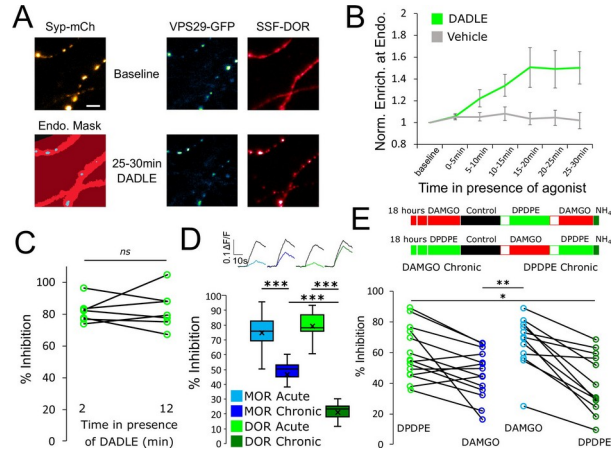
Jullié et al. (2022), Fig. 7.

Delta receptors follow a different trafficking pattern

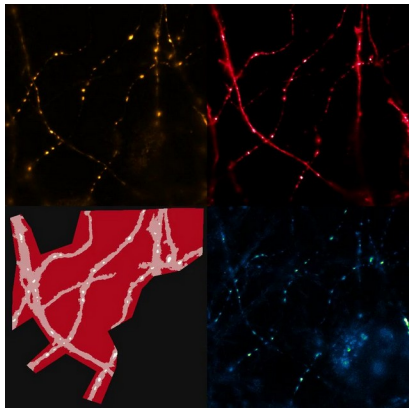
The same axonal study compared delta opioid receptors with mu receptors. Both could be internalized after activation, but their later handling differed. Comparing receptor types in a shared preparation helped identify whether the trafficking effect depended on the receptor rather than the general presence of an axon.

The selected delta-receptor movie follows redistribution after the delta agonist DADLE is added. Receptor fluorescence moves relative to presynaptic and internal-compartment labels. This is the delta-receptor preparation, not another view of the mu-receptor experiment.

Delta receptors recycled less effectively in the tested axons and showed stronger loss of the surface response over prolonged activation. Receptor return to the membrane therefore influenced later signaling. Removing a receptor from the surface and restoring it afterward are separate steps with different functional consequences.



Jullié et al. (2022), Fig. 4.



Delta-receptor trafficking after DADLE. Jullié et al. (2022).

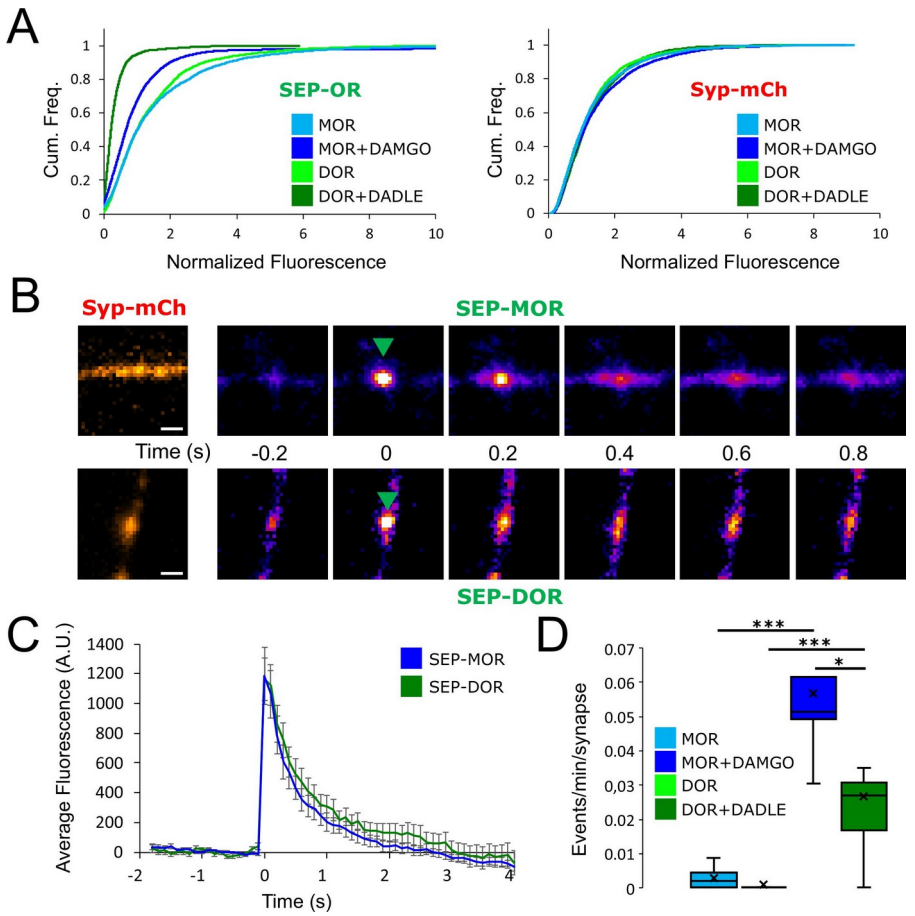
Play video

Endosomes sort receptors after internalization

An endosome is a membrane-bound compartment that receives material taken in from the cell surface. Internalized receptors can be sorted toward recycling routes or toward degradation. The destination determines whether the surface receptor pool can be restored after an agonist-induced uptake event.

Recycling returns receptors to the membrane, where they can again encounter extracellular drug or transmitter. Degradation instead breaks down the receptor and requires replacement by newly made protein. The balance between these routes can change responsiveness during repeated or prolonged exposure.

After internalization, some receptors can continue signaling from inside the cell. Sorting then determines whether their next destination is the surface or a degradative compartment. Recycling restores access to extracellular agonists, whereas degradation removes the receptor protein and makes recovery depend on replacement.



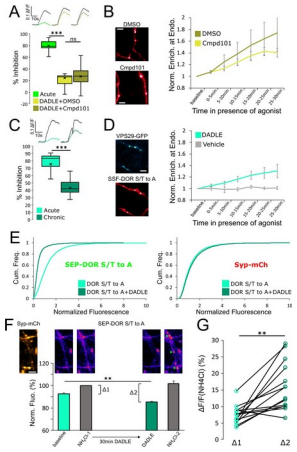
Jullié et al. (2022), Fig. 5.

Phosphorylation changes separate receptor mechanisms

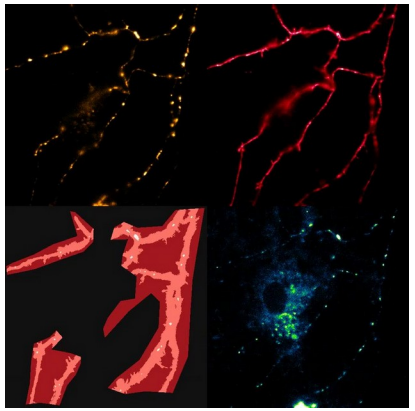
Phosphorylation adds a phosphate group to a protein and can change how other proteins interact with it. The axonal study examined receptor-tail sites that receive these modifications. For mu receptors, GRK2/3 enzymes contributed to the phosphorylation-dependent internalization process.

The selected recording instead follows a delta receptor whose tail serine and threonine sites were replaced. This mutant still underwent agonist-induced internalization in the tested preparation. Delta-receptor uptake therefore did not require the same tail-phosphorylation arrangement as the mu-receptor mechanism.

The comparison separates receptor activation from the machinery that moves the receptor afterward. Two opioid receptor types can inhibit signaling acutely while relying on different rules for internalization and recycling. Those rules help explain why their longer-term axonal tolerance differs.



Jullié et al. (2022), Fig. 6.



Phosphorylation-site mutant delta-receptor trafficking. Jullié et al. (2022).

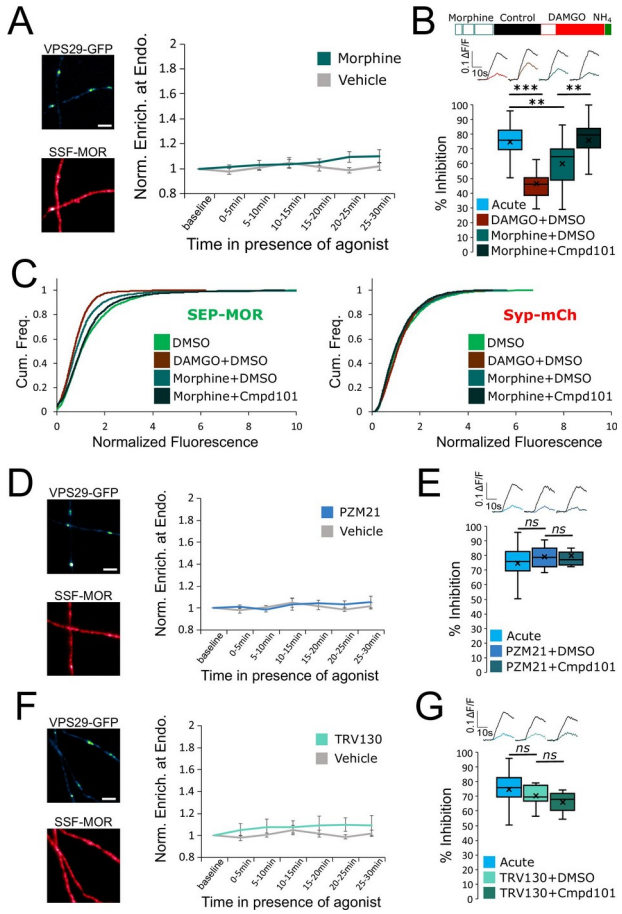
Play video

Receptor regulation depends on the agonist

An agonist activates a receptor after binding, but different agonists can stabilize different receptor states. The resulting balance among G-protein signaling, phosphorylation, and internalization can vary. Equal receptor binding therefore does not guarantee equal trafficking over time.

A partial agonist produces less activation than a full agonist under comparable conditions in a defined system. Its effect also depends on receptor number and the cell's signaling capacity. Partial activation is not the same as failing to bind or having no physiological effect.

Biased signaling describes a preference among a receptor's downstream responses relative to a comparison condition. This must be measured in an actual preparation. A label such as biased does not by itself establish that a drug is free of tolerance, respiratory effects, or misuse potential in people.



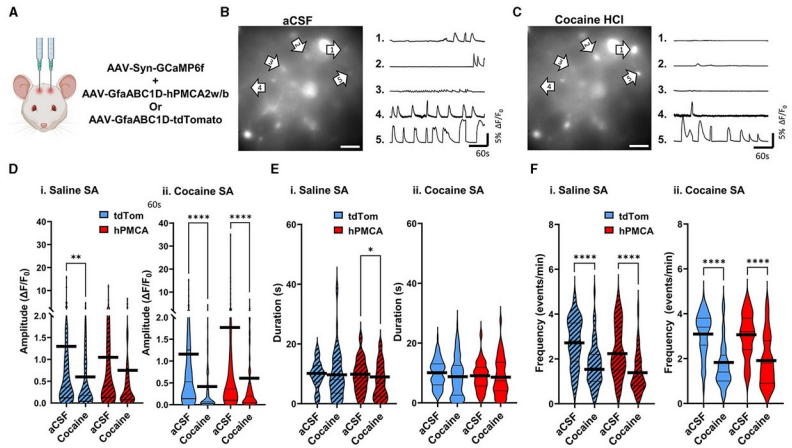
Jullié et al. (2022), Fig. 3.

Baseline imaging follows striatal neuronal calcium

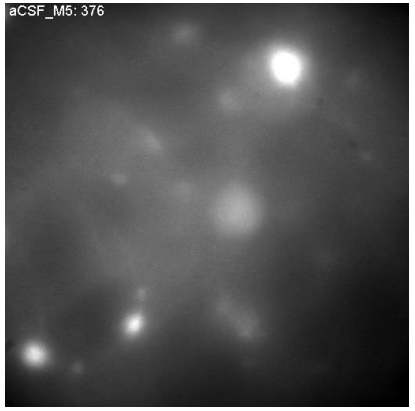
A rat study examined how astrocytes regulate neurons in dorsal striatum after cocaine self-administration. Astrocytes are non-neuronal cells that influence conditions around synapses. Researchers altered their calcium regulation and measured neuronal activity in slices obtained after the animals' training.

The selected original recording shows neuronal GCaMP fluorescence in artificial cerebrospinal fluid before acute cocaine is applied. GCaMP becomes brighter when intracellular calcium rises. The changing signal follows calcium events in neurons, rather than directly measuring dopamine release or the animal's motivation.

This baseline field provides the comparison for the same preparation after cocaine exposure. Its position in the experiment is important: the tissue is being recorded outside the animal. Drug seeking was measured separately in the behavioral portion of the study.



Tavakoli et al. (2024), Fig. 5.



Striatal neuronal calcium: baseline field. Tavakoli et al. (2024).

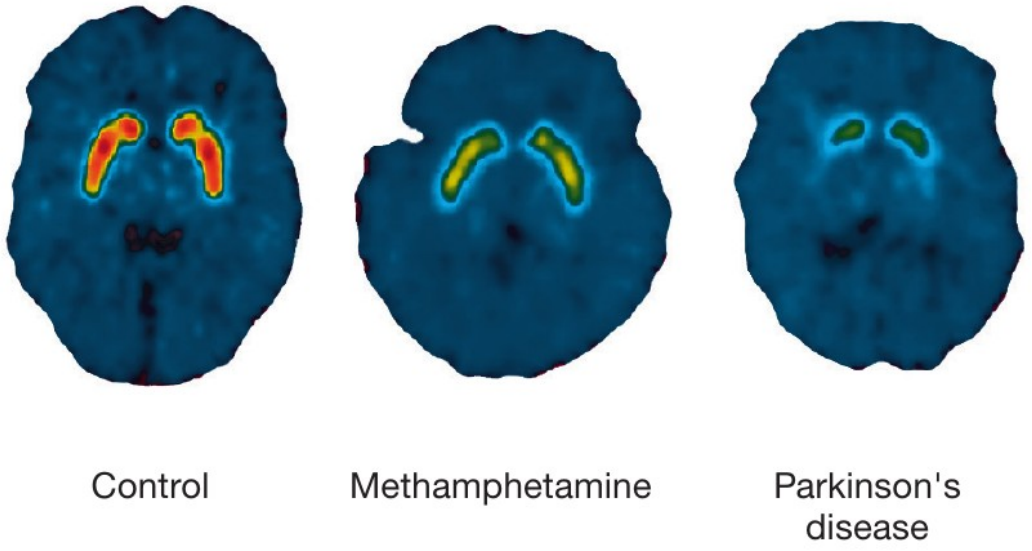
Play video

Cocaine and amphetamine alter dopamine differently

Dopamine transporters normally remove released dopamine from extracellular space by returning it to presynaptic terminals. Cocaine blocks this transporter and prolongs dopamine availability after release. It also affects other monoamine transporters, contributing to effects beyond one dopamine pathway.

Amphetamine enters monoamine terminals and changes the handling of stored transmitter. Its actions can increase cytosolic dopamine and promote outward transport through the dopamine transporter. This differs from simply blocking reuptake while leaving all intracellular transmitter handling unchanged.

Both drugs can increase extracellular dopamine, but their cellular routes and time courses differ. Repeated exposure can also alter learning and circuit regulation. The shared increase in a transmitter does not erase distinctions in the initial pharmacology or later physiological consequences.



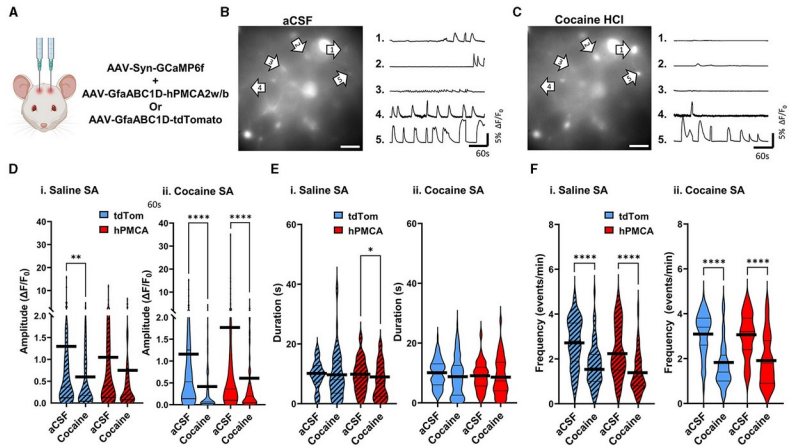
Carlson and Birkett (2022), Fig. 19.14, p.621.

Acute cocaine changes neuronal calcium in the same field

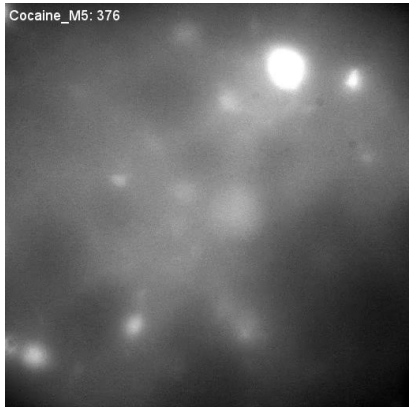
The paired original recording follows the neuronal field after cocaine is applied to the slice. Comparing it with the preceding baseline avoids attributing differences between unrelated fields to the acute drug. The movie reports calcium fluorescence, with the source's accelerated playback preserved.

In the study, acute cocaine reduced neuronal calcium-event amplitude across the tested groups. Prior self-administration changed other aspects of astrocyte control over neurons, but did not abolish this acute response. Acute drug action and experience-dependent regulation were therefore measured separately.

Reducing astrocyte calcium increased cue-induced reinstatement of cocaine seeking in the behavioral experiment. It also increased neuronal-event amplitude specifically after cocaine self-administration. These results identify a contribution of astrocyte regulation to later cue responding, beyond the immediate action of cocaine in the slice.



Tavakoli et al. (2024), Fig. 5.



Same neuronal field after acute cocaine. Tavakoli et al. (2024).

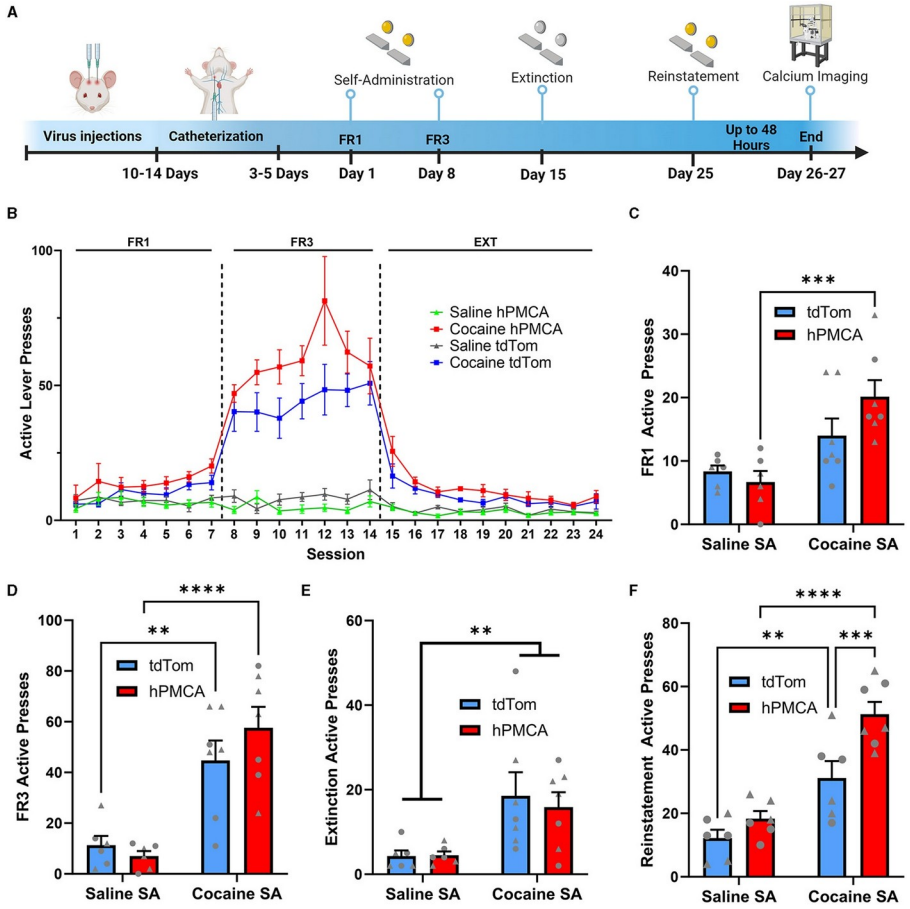
Play video

Drug cues can gain control over action selection

Repeated pairing allows drug-related cues to predict the drug's effects. A cue can then capture attention, increase expectation, or initiate a well-practiced action sequence. Its influence is learned even though the cue itself does not reproduce the drug's full receptor-level effects.

Dorsal-striatal circuits contribute to organizing actions and routines. With repeated experience, behavior can become less dependent on actively evaluating the current outcome at every step. This does not imply that all drug seeking is automatic or that consequences have no influence.

Prefrontal and limbic inputs continue to interact with these action systems. Stress and physiological state can change which signals dominate a choice. Relapse risk therefore concerns both stored cue associations and the current conditions under which those associations are retrieved.



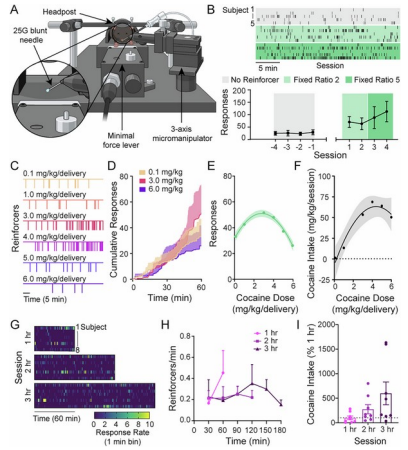
Tavakoli et al. (2024), Fig. 3.

Mice can perform an intranasal self-administration task

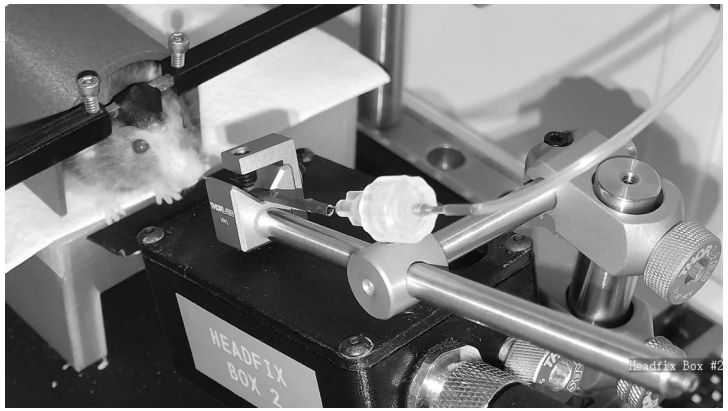
A recent study developed an operant task in which head-restrained male mice obtained intranasal cocaine. The animal's responses controlled delivery under a fixed-ratio schedule. This provided a route-specific preparation for examining voluntary drug taking while maintaining stable access for neural measurements.

The original movie follows responding in the apparatus, with cues marking delivery-related events. It is a task demonstration rather than instructions for human use. The research comparison concerned schedule-dependent responding and verified exposure after the animal earned the outcome.

Self-administration differs from an experimenter giving every animal the same noncontingent treatment. In the operant task, the animal's action changes whether an outcome occurs. This contingency permits reinforcement and drug-related cues to be studied together in a controlled preparation.



Erickson et al. (2025), Fig. 1.



Intranasal self-administration task. Erickson et al. (2025).

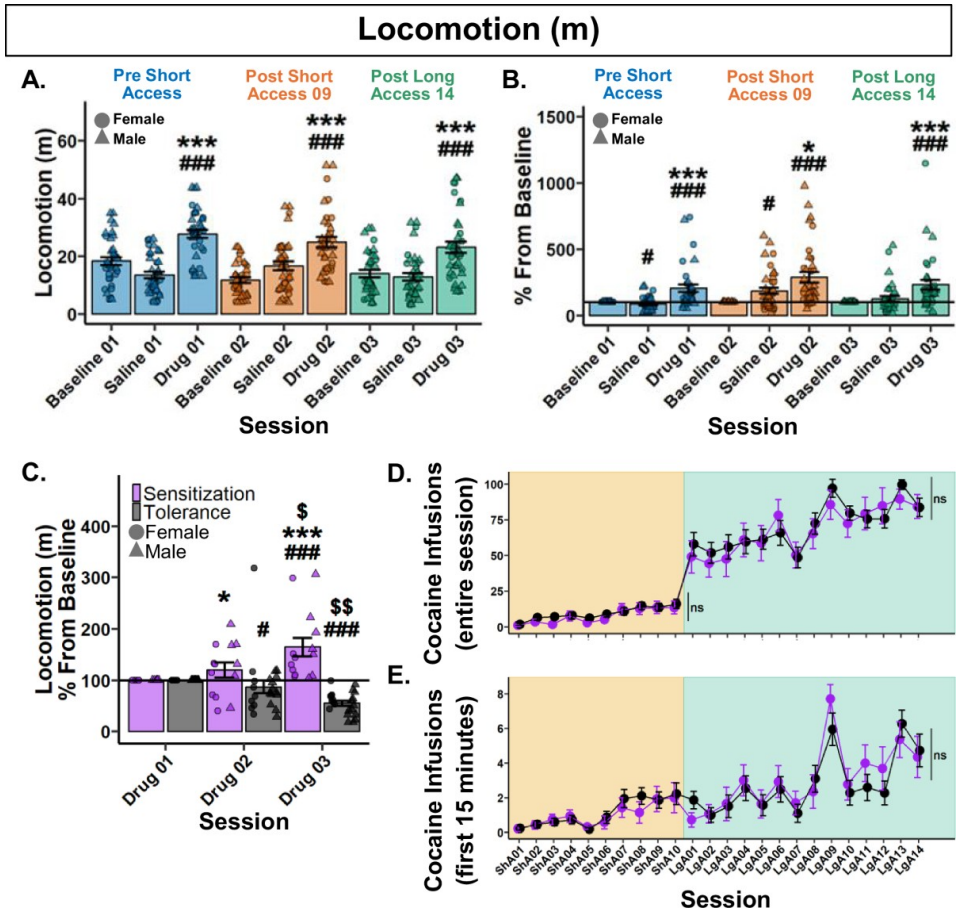
Play video

Sensitization and craving are different outcomes

Sensitization means that a response becomes larger after repeated exposure. Psychomotor sensitization concerns changes in movement-related effects of a drug. Tolerance is a reduction in a response under comparable conditions, so the same exposure history can affect different outcomes in different directions.

Incentive salience concerns the ability of a cue or expected outcome to attract and energize behavior. It is related to wanting rather than necessarily to pleasure. A larger motor response after a drug does not directly establish stronger cue-triggered motivation before the drug arrives.

Experiments must therefore measure movement, cue-directed approach, and intake separately. Their correlations can change across individuals and stages of exposure. A behavioral marker predicts escalation only if its relationship with later intake is tested rather than assumed from the size of an acute drug response.



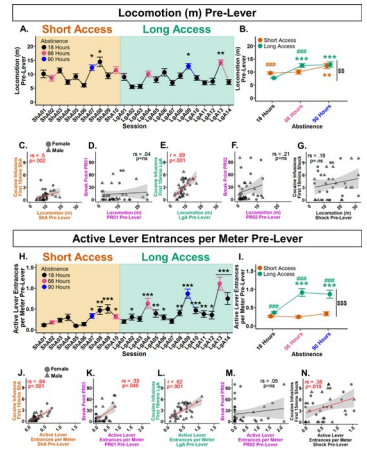
Ramborger et al. (2026), Fig. 2.

Pre-session cue responses predicted intake escalation

Researchers followed cocaine self-administration in genetically heterogeneous rats and used pose-estimation software to track movement. They compared psychomotor sensitization or tolerance with behavior before drug access. This separated drug-induced movement from approach in anticipation of the session.

Pre-session locomotion and approaches to the active-lever area predicted later escalation more effectively than the measured sensitization or tolerance. The findings concerned incentive-related responding before intake. The selected SLEAP recording demonstrates tracking in representative sensitized and tolerant animals, not a direct readout of subjective craving.

Some initially lower-responding rats developed stronger anticipatory behavior and increased intake with extended experience. Vulnerability therefore changed across exposure rather than remaining fixed by the earliest response. The study's longitudinal measures distinguished a developing motivational pattern from one isolated motor effect.



Ramborger et al. (2026), Fig. 3.



SLEAP tracking in sensitized and tolerant rats. Ramborger et al. (2026).

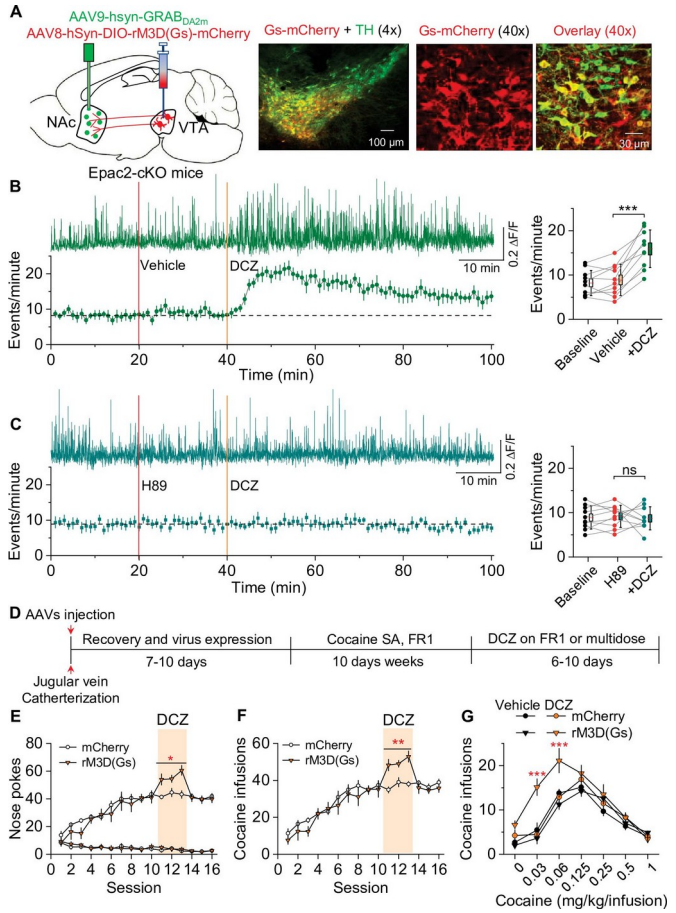
Play video

Epac2 regulates cocaine reinforcement

Cyclic AMP is a messenger inside cells that changes the activity of downstream proteins. Epac2 is one protein activated by cyclic AMP in neurons. Researchers tested whether Epac2 in midbrain dopamine cells contributes to cocaine reinforcement by altering dopamine release.

Removing Epac2 from dopamine neurons reduced evoked dopamine release in nucleus accumbens and reduced cocaine self-administration. A selective Epac2 inhibitor also reduced responding in the tested tasks. These interventions acted on cellular regulation rather than directly blocking cocaine's binding to the dopamine transporter.

Artificially increasing dopamine-neuron activity restored release and reversed the self-administration deficit in the conditional knockout. This rescue connected the behavioral effect to dopamine output. The experiment identifies a candidate cellular mechanism in mice rather than an established clinical Epac2 treatment for cocaine use disorder.



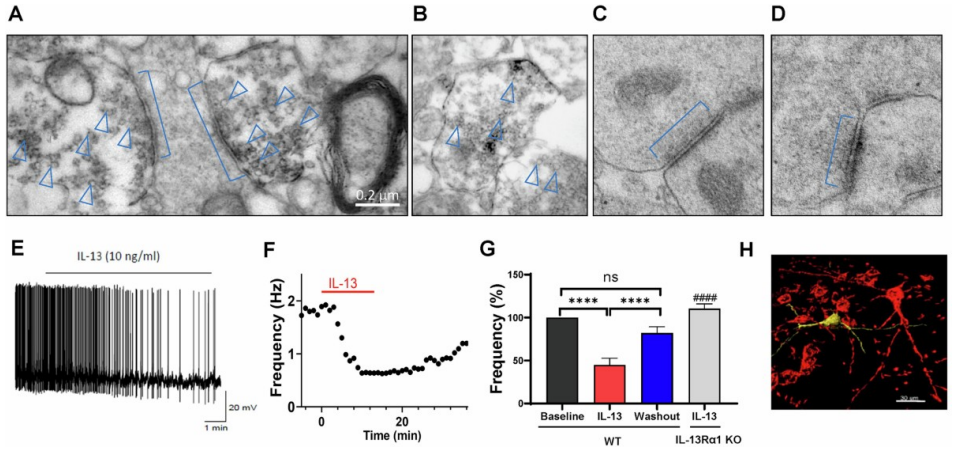
Liu et al. (2022), Fig. 6.

IL-13 can suppress VTA dopamine activity

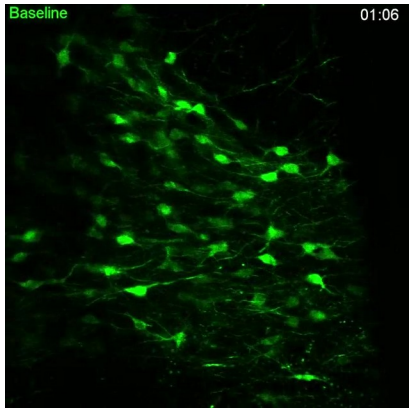
Interleukin-13, or IL-13, is a cytokine that can also be produced by cells in the nervous system. Researchers found its receptor component IL-13R alpha-1 on VTA dopamine neurons. They tested whether this signal changes the cells' activity and their response to nicotine.

Applying IL-13 reduced spontaneous dopamine-neuron activity. The selected calcium recording follows labeled VTA dopamine cells during the treatment. Its fluorescence reports changes in cellular calcium, while separate electrical recordings measured changes in firing.

The effect required the relevant receptor and intracellular signaling involving PI3K and AKT enzymes. Functional HCN channels were also required for the inhibitory response. These experiments identify a specific neuroimmune influence on dopamine cells, rather than treating every cytokine as a uniform source of excitation.



Liu et al. (2025), Fig. 2.



VTA dopamine-cell calcium during IL-13. Liu et al. (2025).

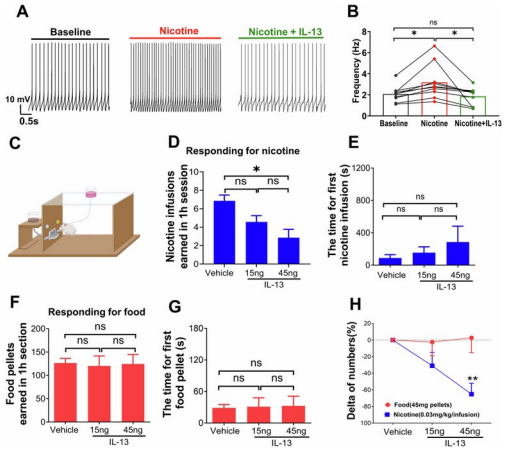
Play video

IL-13 opposes nicotine's stimulation of dopamine cells

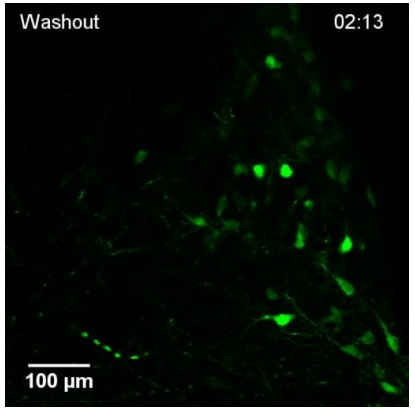
Nicotine increased activity in the tested VTA dopamine preparation, while IL-13 opposed that response. The original second recording follows the treatment comparison with nicotine and IL-13. It shows the cellular response under those conditions rather than the behavior of an animal seeking nicotine.

The study combined calcium imaging with electrical recordings and receptor-related manipulations. Removing the relevant IL-13 receptor component prevented the expected inhibitory action. This distinguished an identified receptor mechanism from a nonspecific reduction in activity caused by adding any protein to the preparation.

Delivering IL-13 locally to VTA also reduced nicotine self-administration in the rodent experiments. The behavioral result followed a targeted brain manipulation, not a clinical trial of systemic cytokine treatment. The cellular and behavioral findings describe different measured stages of the proposed mechanism.



Liu et al. (2025), Fig. 3.



Nicotine and IL-13 treatment comparison. Liu et al. (2025).

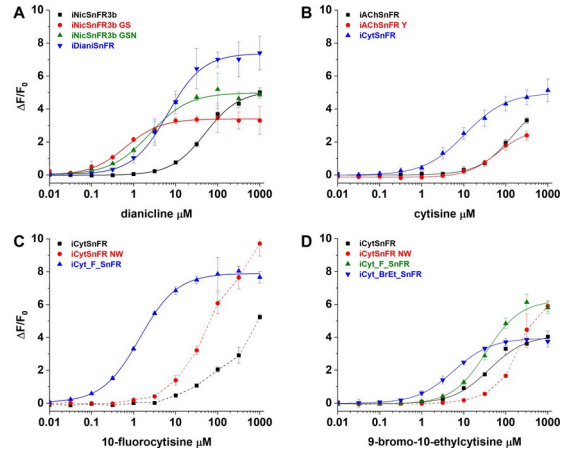
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Biosensors reveal where drug molecules arrive

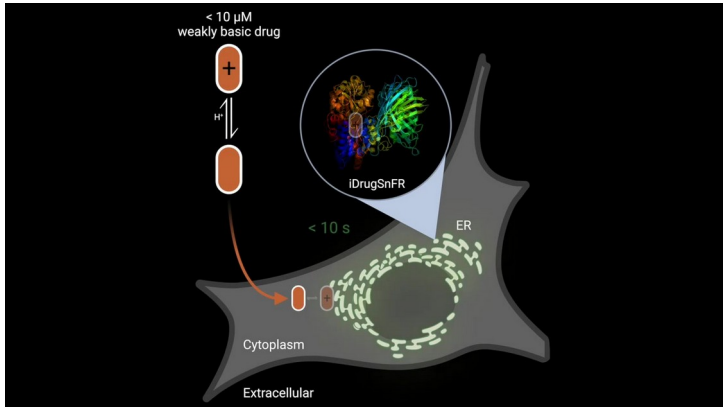
Researchers engineered fluorescent sensors for compounds that act at nicotinic receptors. The sensors combine a drug-binding protein with a fluorescent protein whose brightness changes after binding. They report the presence of the compound rather than the activation of a neuronal nicotinic receptor.

The original introductory movie explains this binding-dependent sensor response. Different versions could be directed to the cell surface or the endoplasmic reticulum, an internal membrane network. Their locations allowed the researchers to compare drug arrival outside the cell with access to an internal compartment.

The study examined partial agonists including dianicline and cytisine. These compounds differed in how quickly they entered the intracellular compartment. A drug can therefore bind a target in one location before reaching a similarly sensitive site behind an additional membrane barrier.



Nichols et al. (2022), Fig. 2.



Fluorescent drug-sensor principle. Nichols et al. (2022).

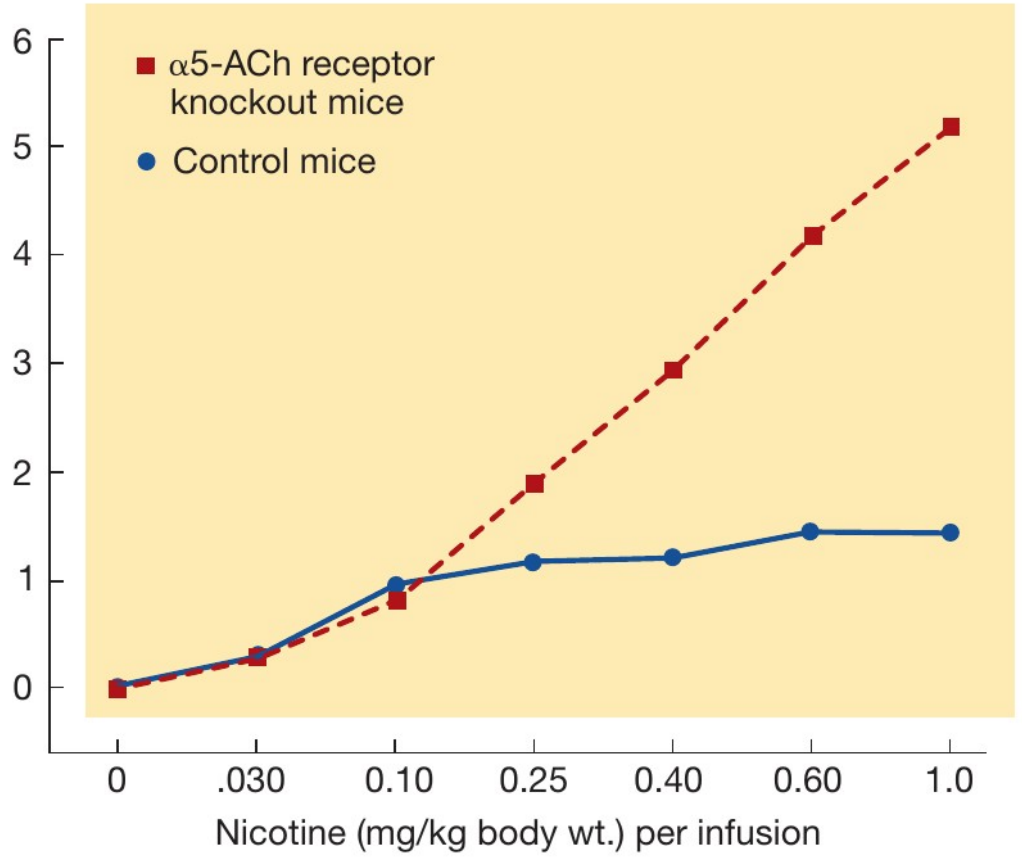
Play video

Nicotine opens nicotinic acetylcholine receptors

Nicotinic acetylcholine receptors are ion channels normally activated by acetylcholine. Nicotine can bind to them and promote channel opening, allowing positively charged ions to enter. The resulting depolarization changes activity in cells and terminals that express suitable receptor combinations.

In VTA circuits, nicotine can influence dopamine neurons directly and through their inputs. Receptors on different cell types can respond and desensitize on different timescales. The resulting dopamine output therefore reflects several cellular actions rather than one uniformly activated population.

Receptor subunits alter channel properties and drug sensitivity. Genetic variation in subunit-related genes can influence nicotine responses and risk, including pathways involving the habenula. The habenula participates in aversive regulation, adding another influence on intake beyond dopamine-related reinforcement.



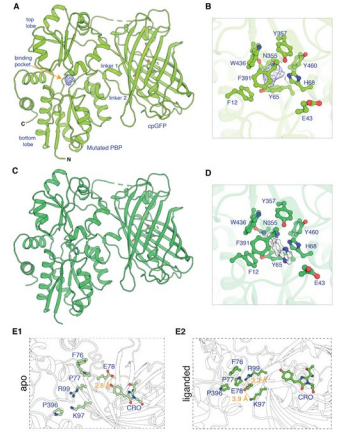
Carlson and Birkett (2022), Fig. 19.17, p.623.

The sensor changes shape when its ligand binds

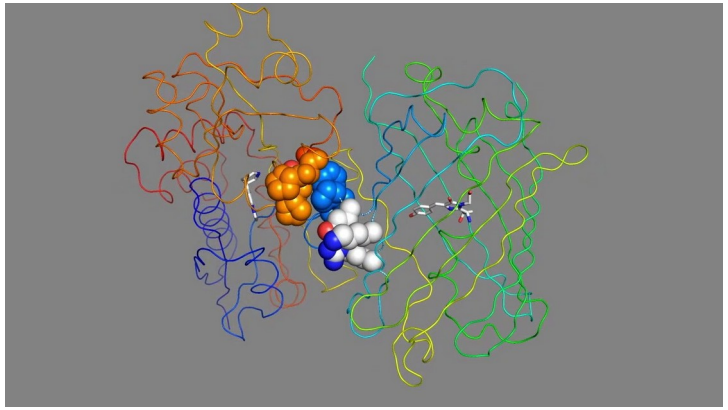
The biosensor study determined structures of its engineered drug-binding protein. A bound ligand stabilizes a different arrangement of the protein's lobes. This movement changes the environment around the attached fluorescent component and provides the basis for a measurable brightness response.

The original structural movie compares sensor conformations with and without bound ligand. It is a published molecular visualization based on the sensor structures. It does not depict a nicotinic acetylcholine receptor opening or ions moving through a neuronal membrane channel.

Separating sensor recognition from receptor activation is essential for interpreting the assay. A fluorescence increase establishes that the sensor encountered a recognized compound in that compartment. The compound's subsequent effect on a natural receptor requires a separate pharmacological or physiological measurement.



Nichols et al. (2022), Fig. 1.



Published sensor-conformation visualization. Nichols et al. (2022).

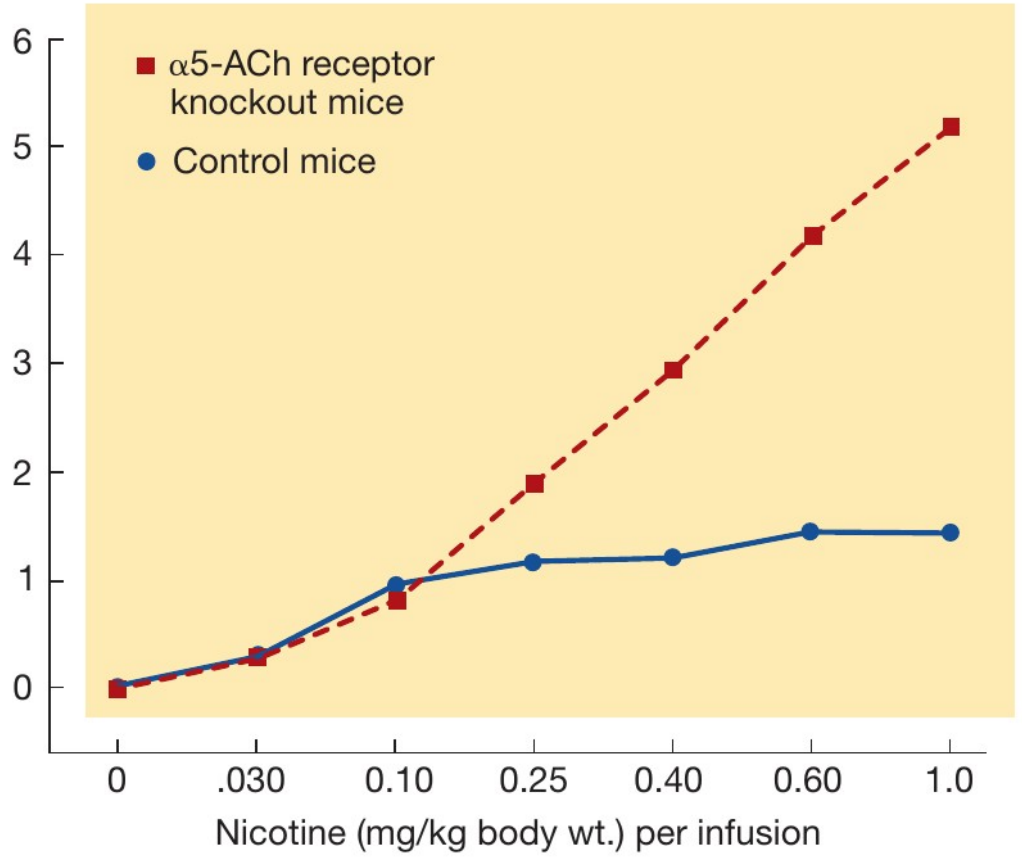
Play video

Genetic risk changes sensitivity rather than destiny

Genes can influence receptor composition, transmitter regulation, and the metabolism of nicotine. A variant can change the probability of developing persistent use without determining the outcome for every carrier. Access, social conditions, stress, and prior experience also influence the eventual pattern.

A receptor's subunits help determine how strongly it responds and how quickly it desensitizes. Variation affecting an aversive response can also change the balance between rewarding and unpleasant drug effects. Risk is therefore not limited to making a reward signal uniformly larger.

Gene-deletion experiments isolate particular mechanisms in a defined animal. Their results can reveal how a receptor contributes to a measured action. They do not establish that the same deletion is the ordinary cause of human tobacco use disorder or that human behavior follows an identical sequence.



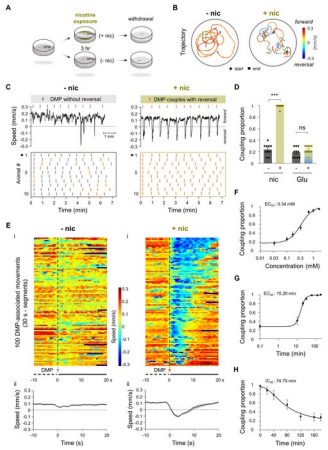
Carlson and Birkett (2022), Fig. 19.17, p.623.

Nicotine can couple normally separate motor programs

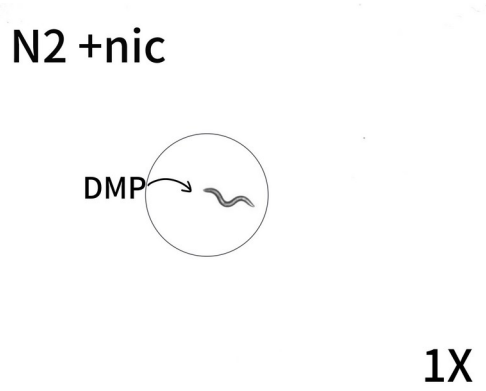
A study in the nematode Caenorhabditis elegans examined how chronic nicotine changes coordination between behaviors. The worm normally produces rhythmic body contractions during its defecation motor program. Researchers found that nicotine increased backward movements and coupled them to these contractions.

The original wild-type recording follows repeated cycles during nicotine exposure. The visible body contraction and reversal are different actions whose timing can be compared. The experiment concerns coordination of a worm's motor programs, not a miniature model of every feature of human addiction.

The investigators identified a mechanosensory pathway from touch-receptor neurons to the AVA interneuron. Nicotine increased the pathway's responsiveness through an ACR-16 nicotinic receptor mechanism. Signals generated by the body's own contraction could consequently trigger an additional movement response.



Liu et al. (2025), Fig. 1.



Wild-type worm motor cycles during nicotine. Liu et al. (2025).

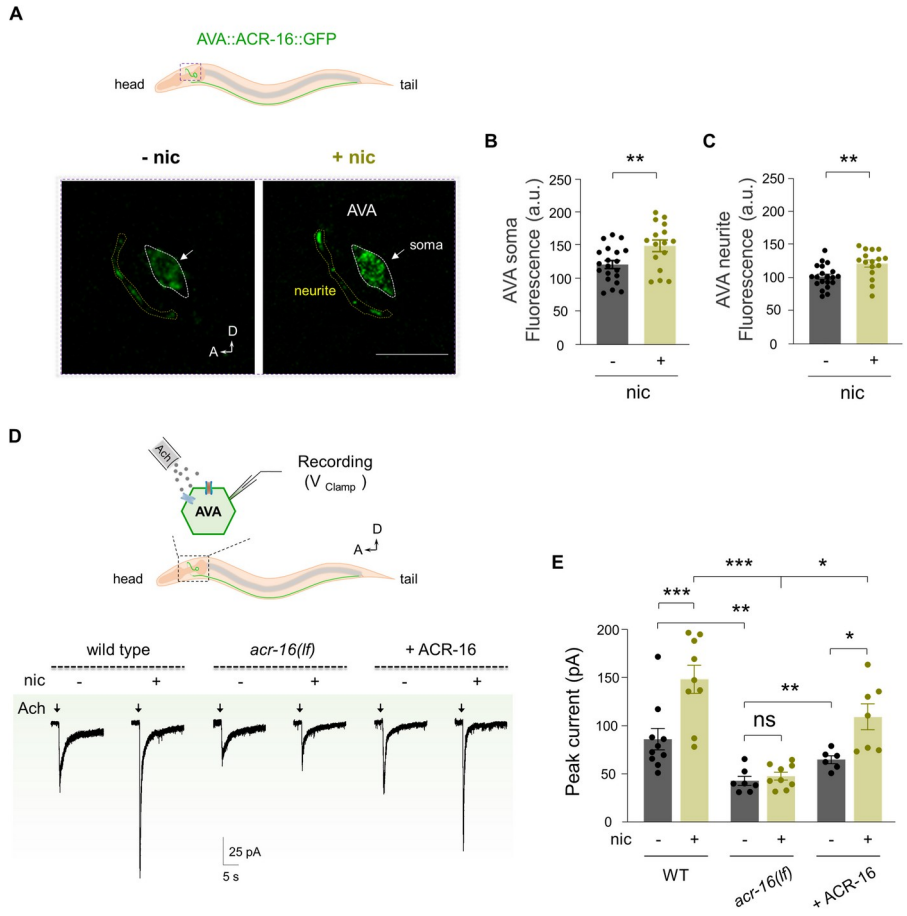
Play video

Desensitization can occur while nicotine remains present

A nicotinic receptor can become less responsive during sustained agonist exposure even while nicotine remains available. This desensitized state differs from an empty receptor that has never encountered the drug. Continued exposure therefore need not produce a constant level of channel opening.

Repeated nicotine exposure can also change receptor abundance and regulation over longer periods. Increased receptor availability and acute desensitization are not contradictory because they describe different processes and timescales. The net physiological response depends on the receptor population and when it is measured.

When nicotine concentrations fall, adapted circuits operate under changed conditions and withdrawal symptoms can emerge. These symptoms can motivate further use through negative reinforcement. The receptor's immediate state and the person's learned response to discomfort contribute at different levels of the maintaining process.



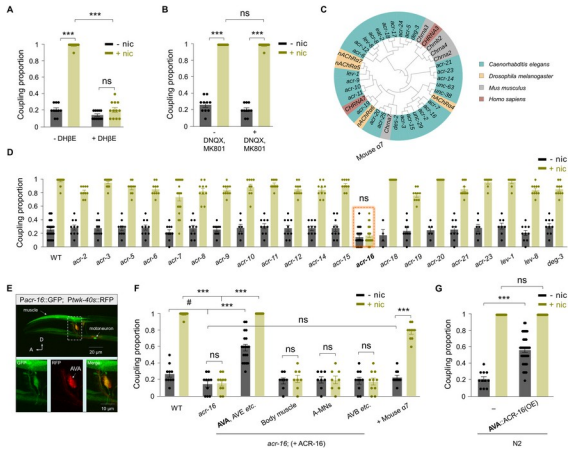
Liu et al. (2025), Fig. 4.

Removing ACR-16 prevents abnormal motor coupling

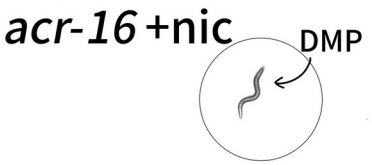
The worm study tested animals lacking ACR-16, a nicotinic receptor subunit required for the identified response. The selected original recording follows these mutants during nicotine exposure. Their contractions can be compared with the additional backward movements seen in nicotine-exposed ordinary worms.

Removing ACR-16 prevented the characteristic nicotine-induced coupling in the tested preparation. Other experiments placed the relevant receptor action in the AVA-related pathway. The mutation therefore interrupted a defined cellular route through which nicotine changed coordination between the motor programs.

Ablating touch-receptor neurons or inhibiting their mechanosensory signaling also removed the coupling. This established that the effect required sensory feedback from the contraction rather than nicotine directly commanding each reversal independently. The pathway transformed an internal body movement into a trigger for another action.



Liu et al. (2025), Fig. 2.



1X

ACR-16 mutant motor cycles during nicotine. Liu et al. (2025).

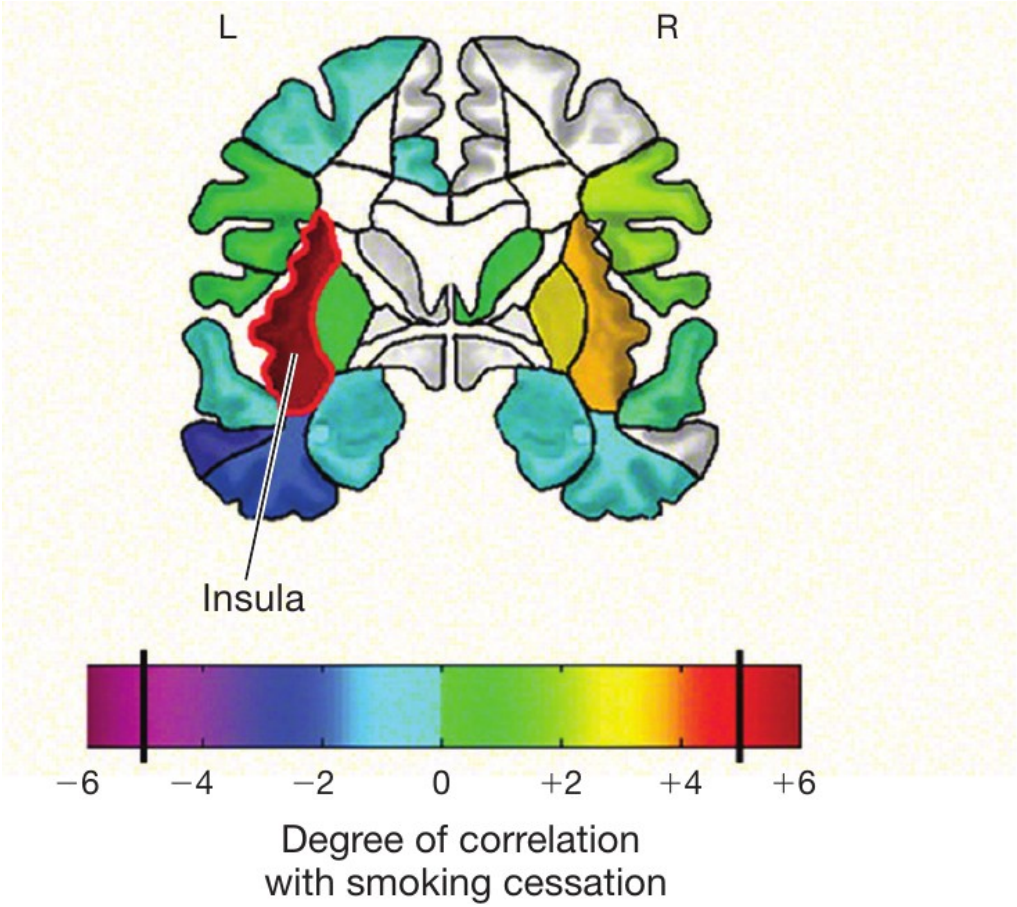
Play video

The insula contributes to craving-related bodily experience

The insular cortex helps represent internal bodily conditions, including sensations associated with drug effects and withdrawal. Such representations can contribute to the felt urge to use. Craving therefore includes bodily experience as well as expectations about an external reward.

Clinical observations of altered smoking behavior after insular injury helped motivate research on this contribution. Injury can affect several functions and connected regions, so the interpretation concerns a network rather than a uniquely isolated craving center. Lesion evidence and experimental stimulation answer different causal questions.

Nicotine-associated cues can retrieve expectations of both bodily relief and reinforcement. Treatment can address these learned responses while also reducing withdrawal through medication. A change in one component does not necessarily remove every cue-triggered urge in all situations.



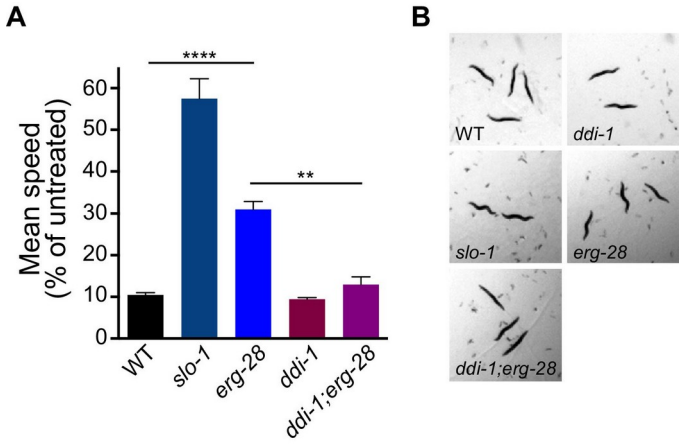
Carlson and Birkett (2022), Fig. 19.18, p.624.

BK-channel regulation changes ethanol sensitivity

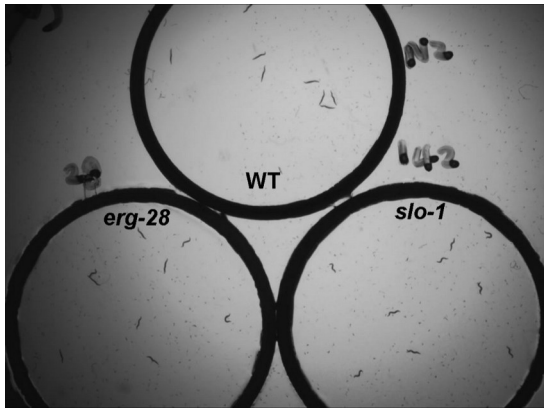
Researchers studied SLO-1, a large-conductance potassium channel in nematodes. Often called a BK channel, it responds to voltage and calcium and helps regulate cellular excitability. The study tested how the protein ERG-28 affects the amount of SLO-1 available at presynaptic sites.

The original behavior recording compares ordinary worms with erg-28 and slo-1 mutants during ethanol exposure. The mutants showed resistance in the tested movement response. This is a locomotor sensitivity assay, not a direct measure of human intoxication, preference, or alcohol use disorder.

Loss of ERG-28 reduced presynaptic SLO-1 because the channel was more vulnerable to degradation before reaching its normal location. A protein involved in handling the channel therefore changed the behavioral effect of ethanol without being the channel through which potassium itself flows.



Oh et al. (2017), Fig. 6.



Ethanol sensitivity in channel-related mutants. Oh et al. (2017).

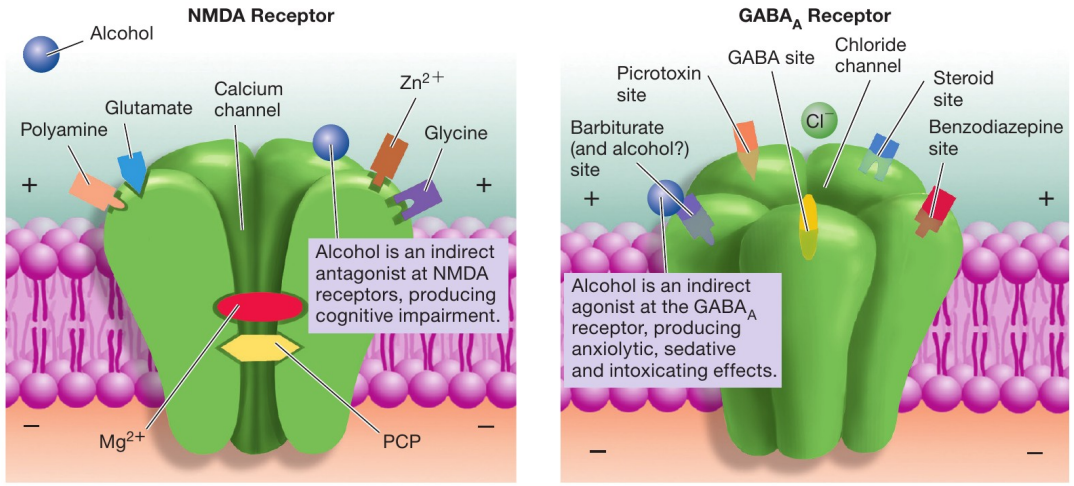
Play video

Alcohol acts at several excitatory and inhibitory targets

Ethanol affects several membrane proteins and cellular processes rather than one exclusive receptor. Its actions can enhance inhibitory signaling and reduce some excitatory transmission. The balance changes with concentration, brain region, and the receptor or channel being examined.

GABA-A receptors mediate an important form of inhibition, while NMDA-type glutamate receptors contribute to excitation and plasticity. Ethanol's effects on these and other targets can impair coordination, memory, and judgment. The overall behavioral state emerges from changes across interacting circuits.

Repeated exposure produces adaptations that oppose aspects of the acute effect. When alcohol is removed after sustained heavy use, excitation can become excessive. This rebound helps explain why alcohol withdrawal can be medically dangerous rather than merely an unpleasant return to sobriety.



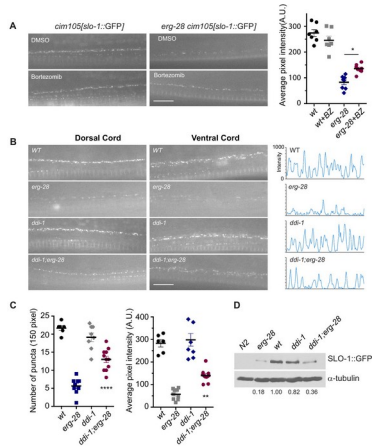
Carlson and Birkett (2022), Fig. 19.20, p.626.

ERG-28 helps a potassium channel reach its destination

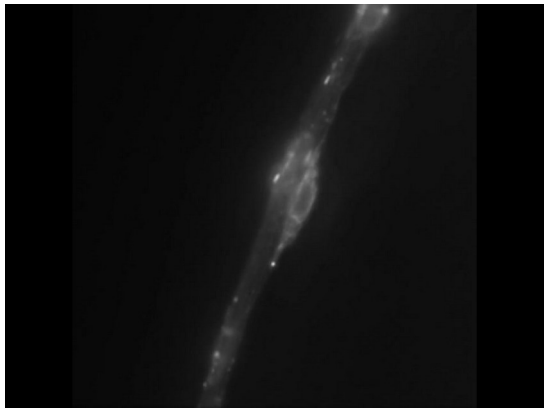
The ERG-28 study followed the protein within cells using a fluorescent tag. The original recording shows mobile ERG-28-associated structures. Their movement provides an observation of intracellular transport, separate from the worm's whole-body movement during the ethanol assay.

ERG-28 protected SLO-1 from degradation involving the DDI-1 protease. A protease is an enzyme that cuts proteins, making them susceptible to removal or changing their function. Protecting the channel helped preserve the pool available for transport toward presynaptic membranes.

Genetic comparisons involving DDI-1 helped identify this degradation step. The resulting channel abundance changed synaptic regulation and ethanol sensitivity. The mechanism concerns protein handling before the channel reaches its functional destination, rather than ERG-28 serving as an ethanol receptor itself.



Oh et al. (2017), Fig. 5.



Fluorescent ERG-28 intracellular movement. Oh et al. (2017).

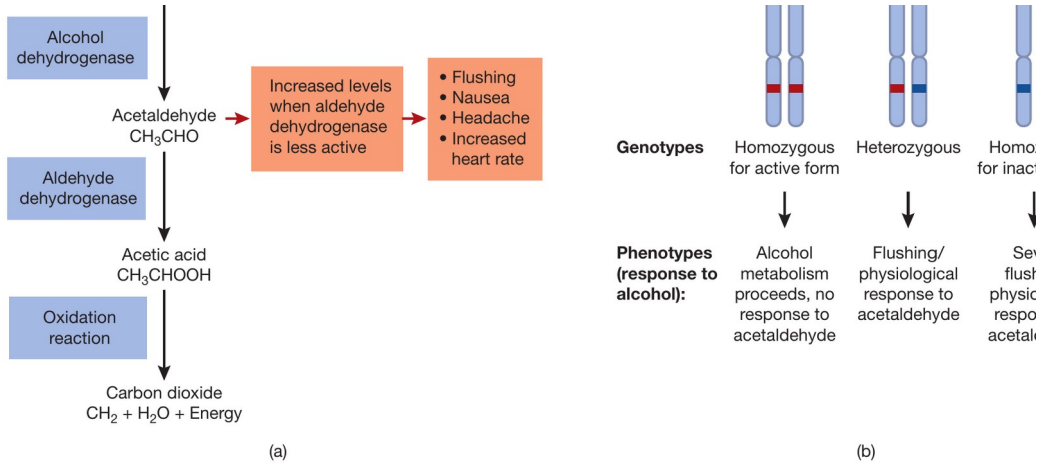
Play video

Alcohol metabolism can generate an aversive intermediate

Alcohol dehydrogenase converts ethanol to acetaldehyde, and aldehyde dehydrogenase helps convert acetaldehyde onward to acetate. Acetaldehyde accumulation contributes to unpleasant reactions such as flushing and nausea. Metabolism therefore changes the bodily consequences of drinking as well as the duration of ethanol exposure.

Some ALDH2 variants reduce the efficiency of acetaldehyde breakdown. The resulting aversive response can reduce the likelihood of heavy drinking in some populations. It does not make alcohol consumption harmless or eliminate the health consequences of acetaldehyde exposure.

Disulfiram interferes with aldehyde dehydrogenase and can make drinking aversive. Its clinical role depends on an informed treatment plan and adherence. This mechanism differs from directly reducing alcohol's reinforcing effects through an opioid-receptor antagonist.



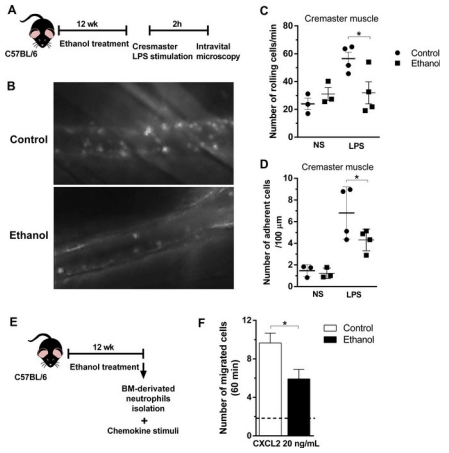
Carlson and Birkett (2022), Fig. 19.11, p.617.

Inflammation recruits leukocytes to a vessel wall

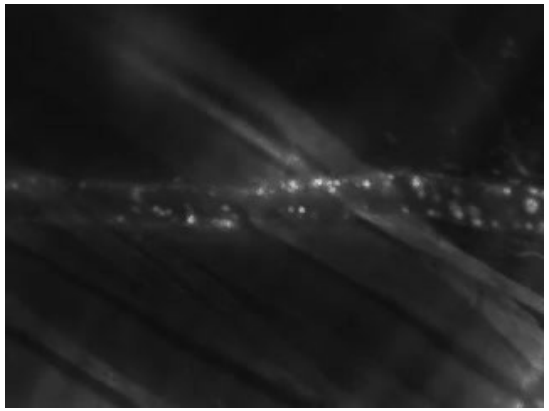
A study of chronic alcohol exposure examined neutrophil responses during inflammation. Neutrophils must interact with the vessel lining before leaving the circulation. The investigators used intravital microscopy, imaging within living tissue, to follow these vascular interactions directly.

The selected control recording follows leukocyte rolling after lipopolysaccharide stimulation. Lipopolysaccharide is a bacterial component that recruits inflammatory responses. The visible moving cells provide a comparison condition for animals with chronic ethanol exposure under the same inflammatory challenge.

Rolling and stable adhesion are distinct stages. A cell can repeatedly contact the vessel wall without making a durable attachment. Measuring these behaviors helped determine whether chronic exposure altered recruitment rather than relying only on the total number of immune cells present in blood.



Malacco et al. (2020), Fig. 6.



Leukocyte rolling: inflammatory control. Malacco et al. (2020).

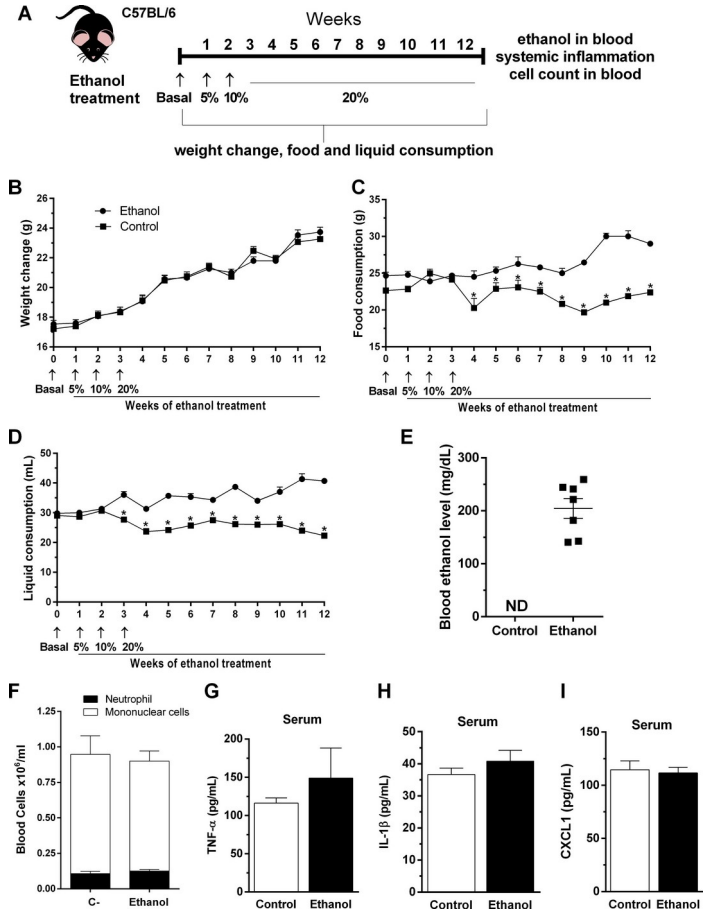
Play video

Alcohol withdrawal can produce excessive excitation

After sustained exposure, neural adaptations can partially counter alcohol's depressant actions. Removing alcohol leaves those adaptations temporarily unopposed. The resulting withdrawal can include tremor, autonomic activation, anxiety, sleep disruption, and, in severe cases, seizures or delirium.

The risk depends on exposure history, previous withdrawal, and medical circumstances. Severe symptoms require medically supervised management. The physiological process differs from a hangover after an isolated episode and should not be treated as the same event.

Repeated withdrawal can also change later responsiveness and vulnerability. Recovery therefore involves stabilization as well as longer-term treatment of learned cues and patterns of use. Suppressing an acute withdrawal syndrome and preventing future harmful drinking are different clinical goals.



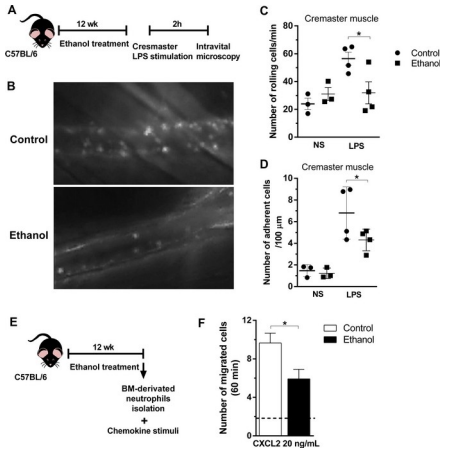
Malacco et al. (2020), Fig. 1.

Chronic ethanol changes inflammatory-cell recruitment

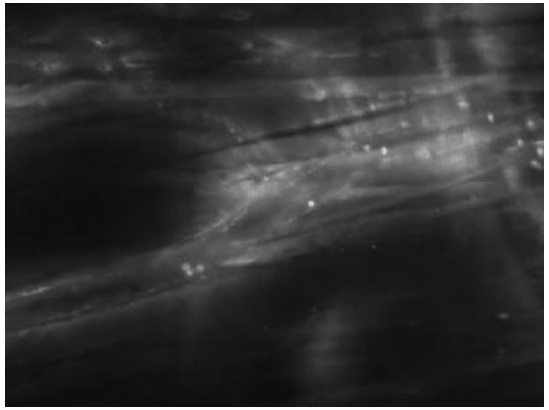
The paired intravital recording follows the inflammatory challenge after chronic ethanol exposure. Its preparation matches the control comparison while differing in exposure history. The moving leukocytes can be followed as they contact and roll along the vessel lining.

The study found impaired neutrophil recruitment and reduced antifungal defense after chronic ethanol exposure. These effects concerned the ability to organize a protective response, not merely a reduction in visible movement during intoxication. The immune measurements were made in defined inflammatory and infection-related preparations.

The paper connected altered recruitment with impaired clearance of Aspergillus in the tested model. Vascular behavior and infection outcome provide different stages of evidence. Chronic alcohol can therefore alter host defense through cellular processes beyond its immediate effects on neural inhibition and coordination.



Malacco et al. (2020), Fig. 6.



Leukocyte rolling after chronic ethanol. Malacco et al. (2020).

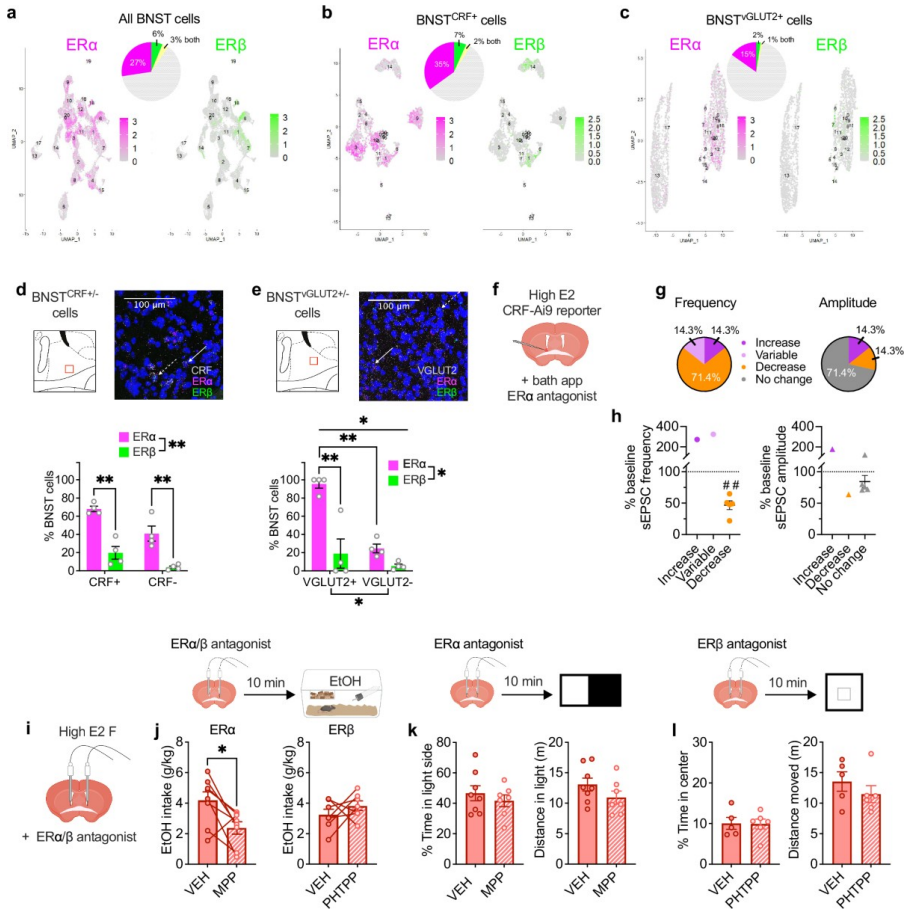
Play video

Membrane estrogen signaling increases alcohol drinking

A mouse study tested how ovarian estrogen changes binge-like alcohol drinking. Female mice drank more when endogenous estrogen was high. The investigators distinguished rapid signaling at membrane-associated receptors from slower changes that require regulation of gene expression.

In the bed nucleus of the stria terminalis, membrane-associated estrogen receptor alpha increased excitation of CRF-containing neurons. CRF is a stress-related peptide, and these cells became more active during drinking. Local manipulations identified this pathway as contributing to the increased intake.

The effect concerned alcohol drinking rather than all anxiety-related measures changing together. The study separated the pro-drinking action from reduced avoidance. A hormone can therefore influence intake through a defined rapid circuit mechanism without every behavioral effect being explained by one change in anxiety.



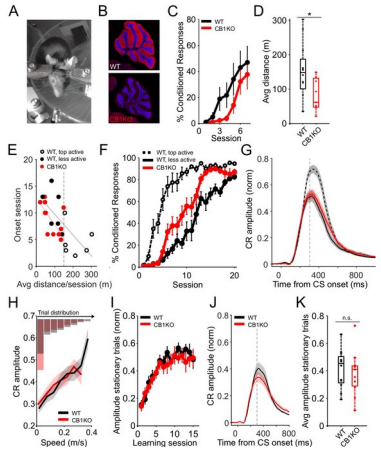
Zallar et al. (2024), Fig. 5.

CB1 deletion changes the state in which learning is tested

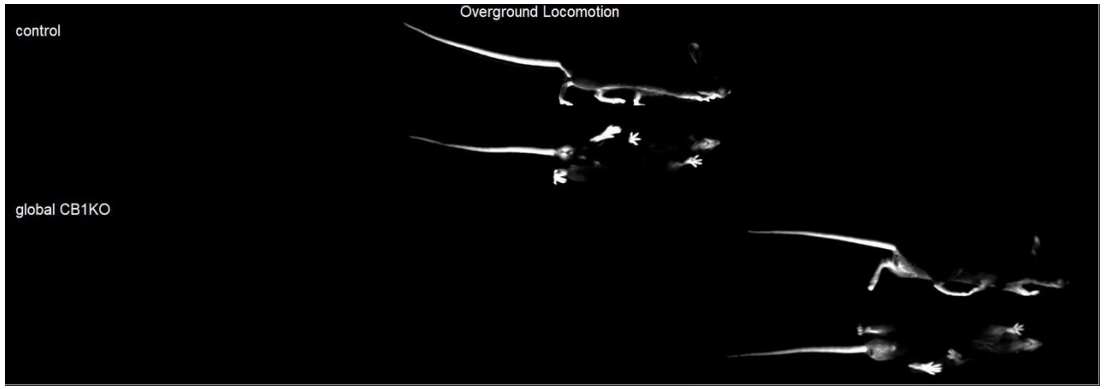
Researchers examined mice lacking CB1 cannabinoid receptors throughout the body. These animals were less active and showed an apparent impairment in delay eyeblink conditioning. The task measures learning to close the eyelid in anticipation of a later stimulus.

The selected original recording compares an ordinary mouse and a global CB1-knockout mouse walking overground. It demonstrates the locomotor preparation used to assess coordination, not the eyelid-conditioning session. The investigators distinguished movement patterns from the reduced spontaneous activity that affected learning conditions.

When training occurred on a motorized treadmill, the apparent conditioning deficit was rescued. The result indicated that behavioral state contributed strongly to the earlier impairment. Removing a receptor could affect the task indirectly through activity rather than solely by deleting a synaptic learning mechanism.



Albergaria et al. (2020), Fig. 1.



Overground walking: control and CB1 knockout. Albergaria et al. (2020).

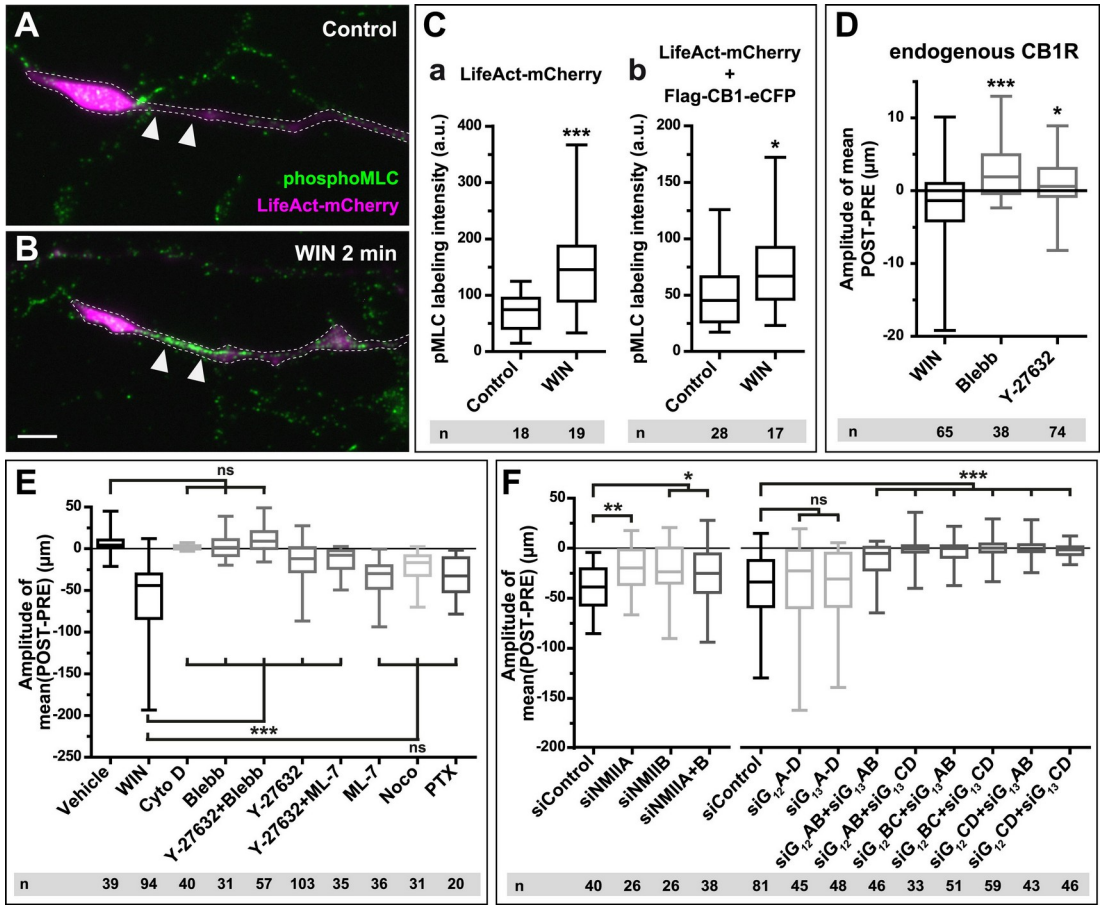
Play video

THC acts at receptors for endocannabinoids

Endocannabinoids are lipid-derived messengers produced by the body. They can travel backward across a synapse from a receiving cell to presynaptic CB1 receptors. This retrograde signal reduces transmitter release and helps regulate how strongly an input influences its target.

Tetrahydrocannabinol, or THC, activates cannabinoid receptors and can alter perception, memory, appetite, and movement. Its externally supplied time course differs from locally produced endocannabinoid signaling. Widespread receptor exposure therefore need not reproduce the precise timing of a normal synaptic feedback signal.

Cannabidiol, or CBD, has a different pharmacological profile involving several targets and modulatory actions. It is not simply a universal CB1 blocker or a guaranteed treatment for psychosis. The effects of a cannabinoid preparation depend on its components, exposure, and the outcome being measured.



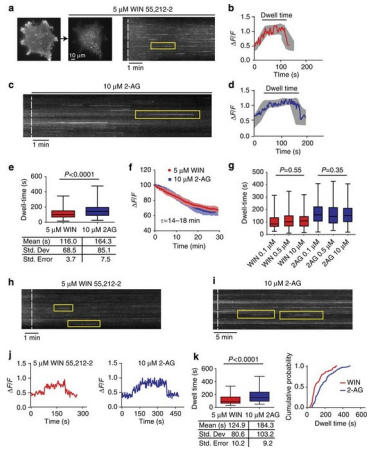
Roland et al. (2014), Fig. 3.

A surface tag distinguishes receptor location

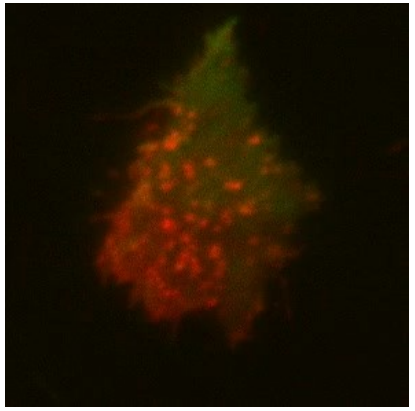
A CB1-receptor study used a fluorescent tag sensitive to acidity to distinguish surface receptors from internalized ones. In cultured cells, the tag was bright at the outer membrane. Briefly acidifying the external solution rapidly quenched fluorescence exposed to that solution.

The selected original recording follows this surface-quenching test together with beta-arrestin labeling. Beta-arrestin is a protein that can bind activated receptors and participate in their internalization. The preparation identifies where receptor-associated fluorescence is located rather than measuring cannabis intoxication.

The experiment used an agonist that activates CB1 receptors and promotes clustering and uptake. Comparing the acidity response with receptor movement helped distinguish surface clusters from compartments inside the cell. This supplied a location test for the later analysis of receptor trafficking and signaling.



Flores-Otero et al. (2014), Fig. 1.



Surface CB1 fluorescence quenching. Flores-Otero et al. (2014).

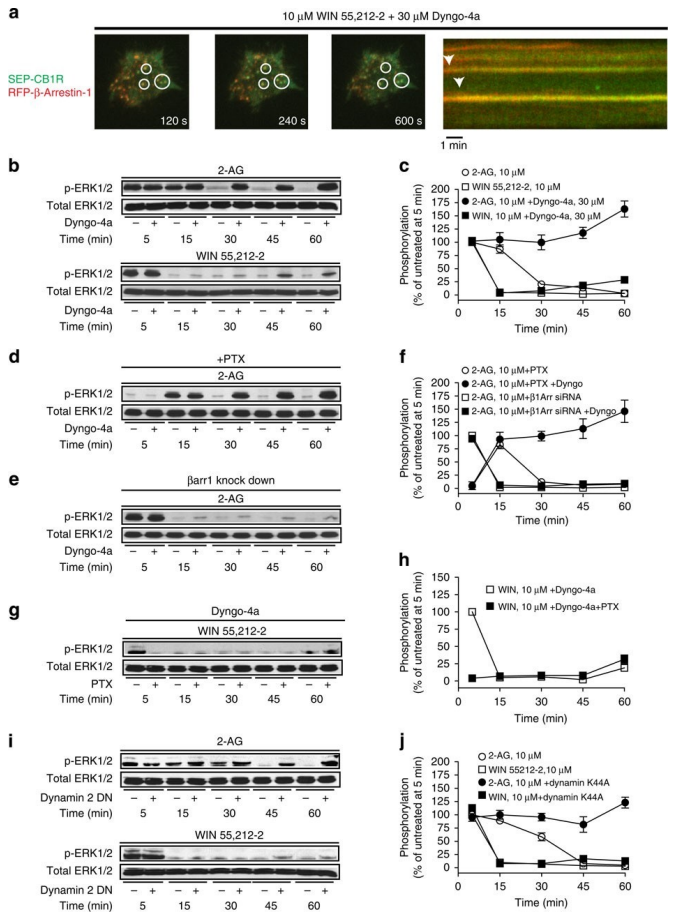
Play video

Different ligands change CB1 endocytic timing

The CB1 study compared how different activating ligands affected receptor clusters at the cell membrane. Receptors spent different amounts of time in endocytic structures before being internalized. Endocytic dwell time therefore depended on the ligand rather than being fixed for every activation event.

The researchers related these timing differences to beta-arrestin recruitment and downstream ERK signaling. ERK is an enzyme pathway that changes other proteins through phosphorylation. Altering the duration of receptor-associated complexes changed the opportunity for signaling from that location.

A receptor's effect therefore depends partly on how long it remains in a signaling compartment. Binding and channel regulation are not the only relevant events after an agonist arrives. Trafficking can change both the location and duration of the signal generated by the same receptor type.



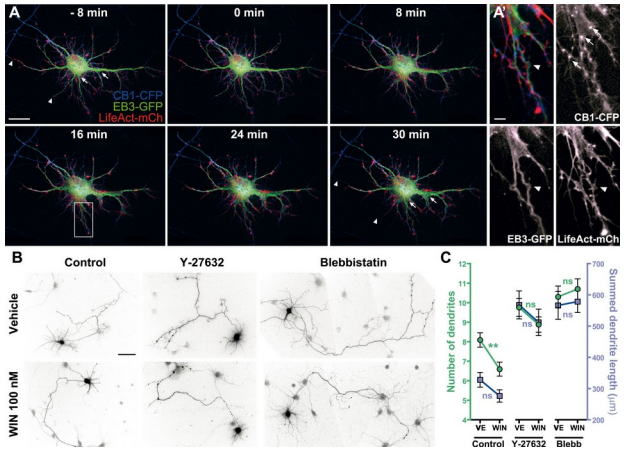
Flores-Otero et al. (2014), Fig. 6.

CB1 activation can change neuronal shape

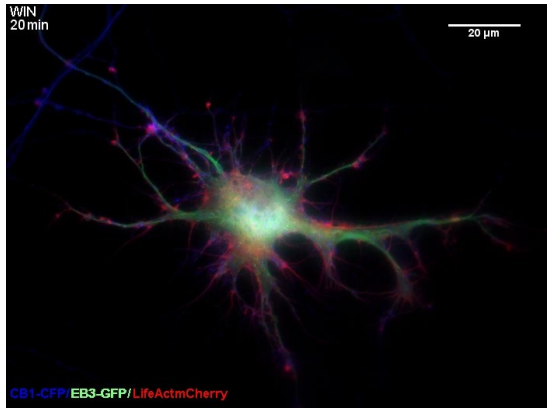
Researchers imaged cultured hippocampal neurons while activating CB1 receptors. The selected recording follows the cell body and dendrites, together called the somatodendritic region. Fluorescent labels identify the receptor and components of the internal framework as the same neuron changes over time.

CB1 activation produced changes in cell shape through signaling that regulates actomyosin. Actin forms structural filaments, while myosin uses chemical energy to generate force against them. Changing their organization can move the membrane and alter the shape of neuronal processes.

The study separated effects on actin from those involving microtubules and contractile machinery using inhibitors. These experiments concerned cultured neurons and an applied agonist, not a demonstration that ordinary cannabis exposure produces an identical structural change throughout an intact adult human brain.



Roland et al. (2014), Fig. 7.



Somatodendritic response to CB1 activation. Roland et al. (2014).

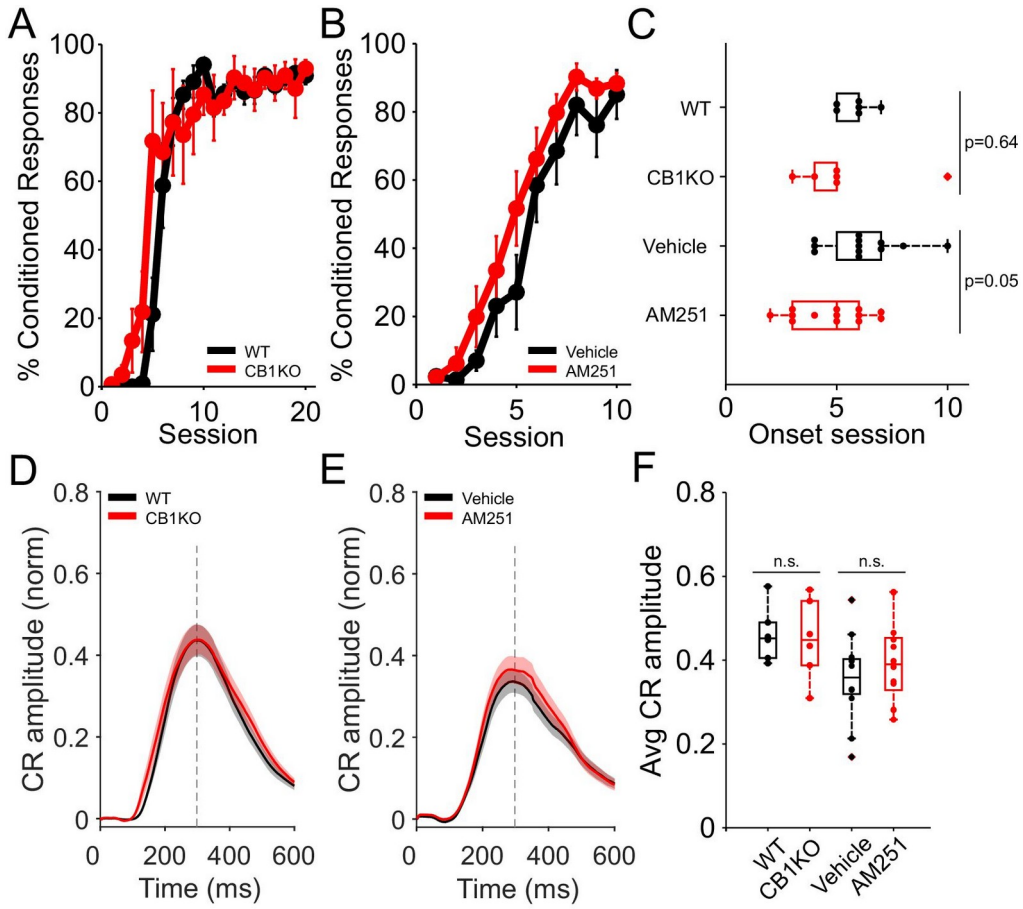
Play video

Activity state can imitate a cannabinoid learning deficit

The CB1-knockout learning study compared animals and trials with different amounts of movement. Lower activity explained much of the apparent conditioning impairment. A gene deletion could therefore change performance by shifting the state in which the learning task was experienced.

Providing movement on a motorized treadmill restored delay eyeblink conditioning in global knockouts. This rescue did not replace the deleted receptor. Instead, it changed a behavioral condition that influenced how effectively the remaining circuitry performed the task.

Deleting CB1 specifically from cerebellar granule cells did not reproduce the global knockout's conditioning deficit. The anatomical specificity further separated a proposed local plasticity mechanism from whole-animal state. Global and cell-restricted interventions therefore answered different questions about the receptor's contribution.



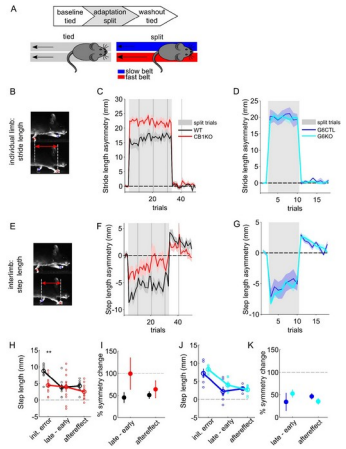
Albergaria et al. (2020), Fig. 2.

Motor coordination can remain intact without global CB1

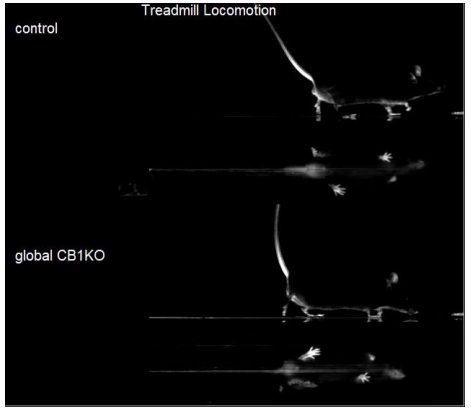
The selected treadmill recording compares control and global CB1-knockout mice during the tied-belt condition. Both sides of the treadmill move together, providing a controlled walking demand. The recording belongs to the locomotor assessment rather than the separate eyeblink-conditioning experiment.

The study found preserved cerebellum-dependent locomotor coordination and learning in the tested knockout groups. Reduced spontaneous activity therefore did not imply a complete inability to coordinate movement. Motivation or state can differ even when a constrained locomotor task remains possible.

Combining the movement tests with conditioning clarified which outcome was impaired under which conditions. A change in one behavioral assay cannot be generalized to every cerebellar function. The receptor's contribution must be assigned to the measured process and the state in which that process was tested.



Albergaria et al. (2020), Fig. 4.



Tied-belt walking: control and CB1 knockout. Albergaria et al. (2020).

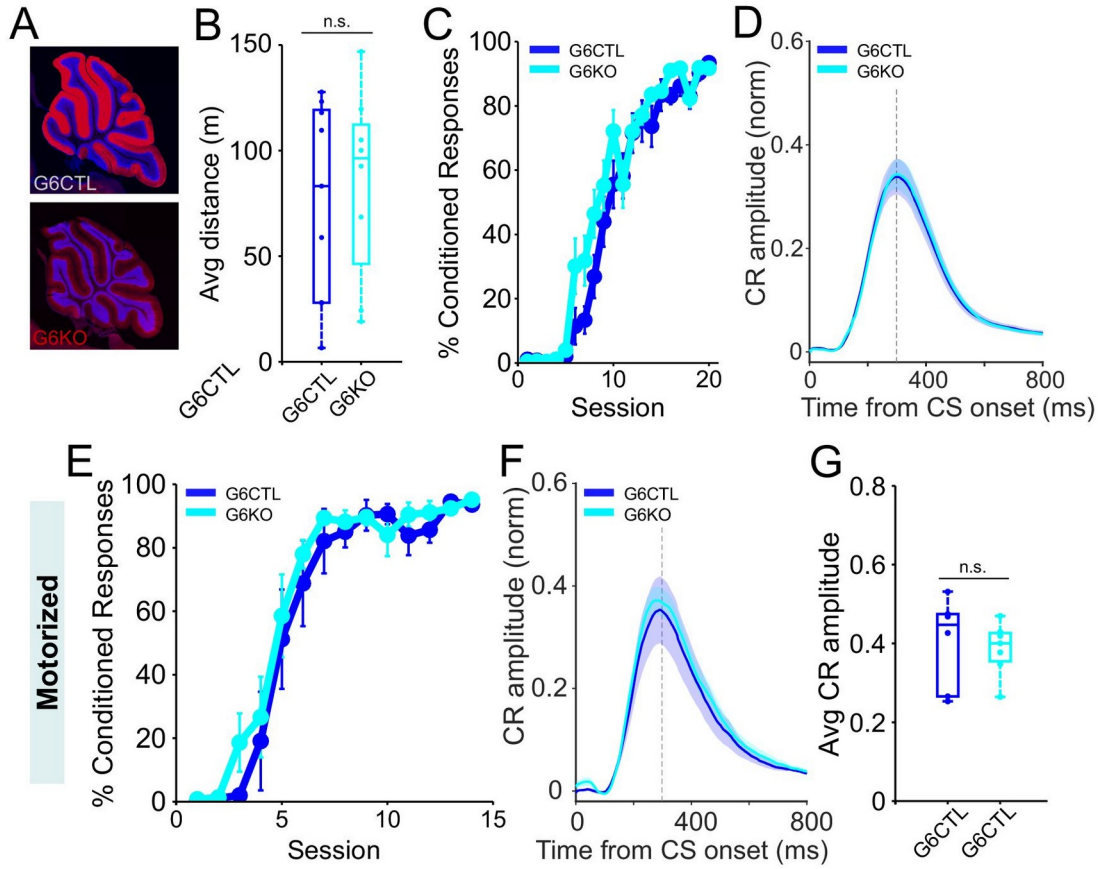
Play video

Local plasticity and whole-animal performance can diverge

CB1 receptors contribute to several forms of synaptic plasticity in cerebellar slices. Plasticity means a change in how effectively one cell influences another. A missing slice-level response can identify a molecular contribution without proving that every behavior thought to use that circuit will fail.

In the learning study, both global and granule-cell-specific receptor deletions were evaluated with several tasks. The results did not support a simple requirement for granule-cell CB1 in all measured learning. The preparation and behavioral state changed the apparent consequences of the deletion.

A rescue through increased movement showed that intact components could support conditioning when the state was changed. This is different from restoring the receptor or repairing every altered synapse. The experiment identified an indirect route from a gene to task performance through ongoing activity.



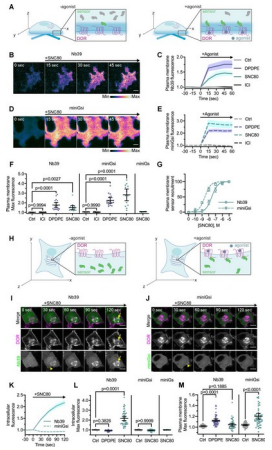
Albergaria et al. (2020), Fig. 3.

Delta receptors can signal from an internal compartment

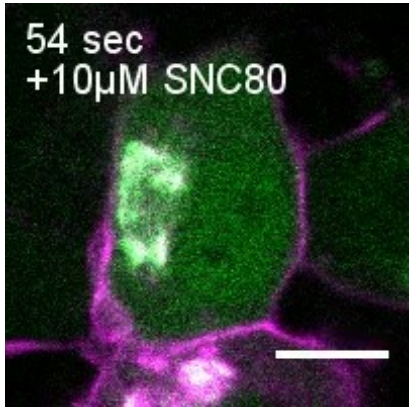
Researchers examined delta opioid receptors in cultured PC12 cells, a laboratory cell line used to study cellular signaling. Some receptors were present in the Golgi apparatus, an internal membrane system that processes and sorts proteins. A membrane-permeant agonist could reach these internal receptors.

The original recording follows recruitment of Nb39, a sensor that recognizes an activated receptor conformation. After the agonist SNC80 was added, the sensor accumulated at the Golgi receptor pool. The fluorescence therefore indicated an activated receptor state at an intracellular location.

This experiment concerns delta receptors in cells rather than mu receptors during opioid-maintenance treatment. Its pharmacological question is access to an internal receptor population. A compound can encounter surface and intracellular targets differently depending on whether it crosses the intervening membranes.



Crilly et al. (2021), Fig. 1.



Nb39 recruitment to Golgi delta receptors. Crilly et al. (2021).

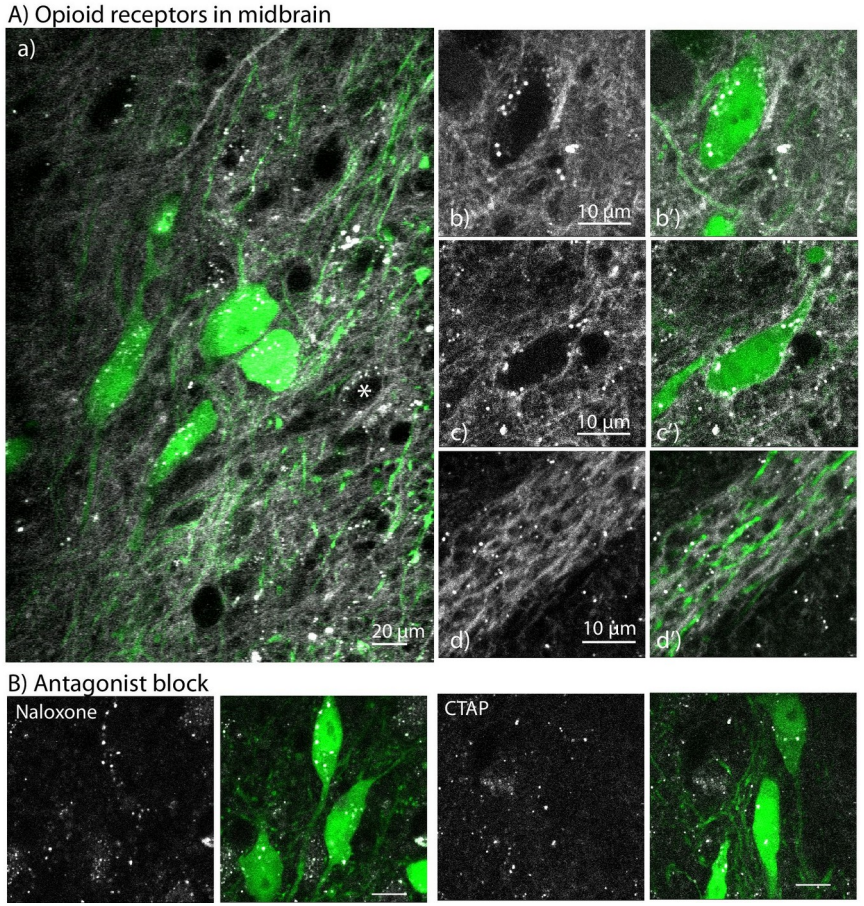
Play video

Maintenance treatment stabilizes opioid-receptor exposure

Methadone is a long-acting mu-opioid-receptor agonist used in supervised treatment of opioid use disorder. Stable exposure can reduce withdrawal and the repeated cycle of short-lived effects followed by distress. Treatment aims to improve health and functioning while reducing harmful patterns of use.

Maintenance is not the same intervention as repeatedly seeking a rapidly changing intoxicating effect. Dose, timing, monitoring, and treatment context alter the pattern of receptor exposure. The medicine still has physiological effects and requires individualized care rather than being assumed harmless.

Psychological and practical support can address cues, stress, health needs, and daily functioning alongside medication. Preventing withdrawal removes one maintaining pressure but does not automatically erase learned associations. Continued treatment can therefore target several processes that contribute to persistent use.



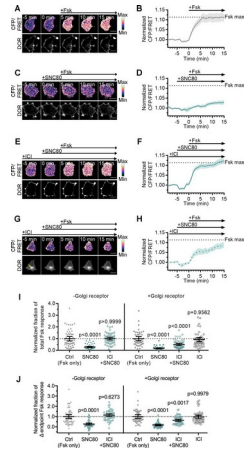
Arttamangkul et al. (2019), Fig. 6.

Two activation sensors report different receptor states

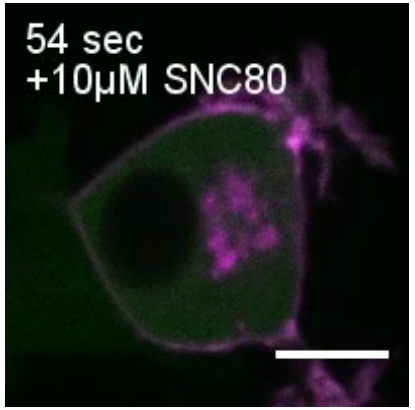
The delta-receptor study compared Nb39 with miniGsi, a protein probe designed to recognize a G-protein-related receptor state. The probes did not behave identically at the Golgi. A receptor could therefore be active according to one molecular readout without recruiting every activation sensor in the same way.

The selected recording shows that miniGsi did not accumulate at the internal Golgi pool after agonist application, unlike Nb39 in the preceding experiment. Separate assays still detected inhibition of cyclic AMP production. Failure of this probe to bind did not mean that all signaling from the receptor was absent.

The internal membrane environment can shape the receptor conformations and partners that are available. Location therefore changes more than physical access to a binding site. It can alter which signaling complex forms after a compound has already reached and activated the receptor.



Crilly et al. (2021), Fig. 4.



miniGsi response at Golgi delta receptors. Crilly et al. (2021).

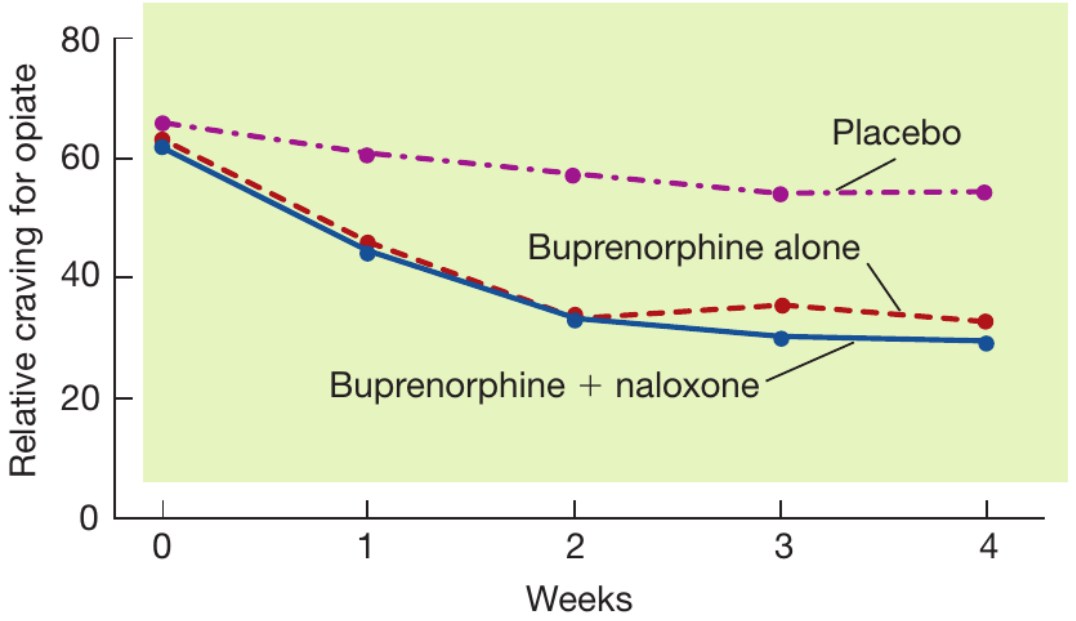
Play video

Buprenorphine and naloxone have different roles

Buprenorphine is a high-affinity partial agonist at mu opioid receptors. It can suppress withdrawal and reduce the effects of other opioids under appropriate treatment conditions. Partial agonism limits some responses relative to a full agonist, but does not remove all risk or misuse potential.

Naloxone is an opioid-receptor antagonist that can reverse opioid-induced respiratory suppression during overdose. It serves a different immediate purpose from ongoing maintenance treatment. Reversal can be temporary relative to the duration of the opioid, so emergency assessment remains essential.

Some buprenorphine formulations include naloxone to discourage particular forms of misuse. That combination is distinct from administering rescue naloxone during an overdose. Treatment planning separates stabilization, prevention of recurrence, and emergency reversal because these interventions address different stages of the problem.



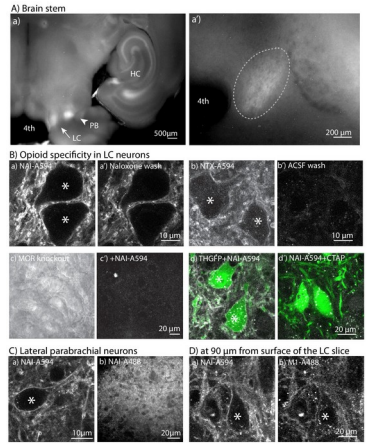
Carlson and Birkett (2022), Fig. 19.25, p.631.

A labeling reaction leaves opioid receptors functional

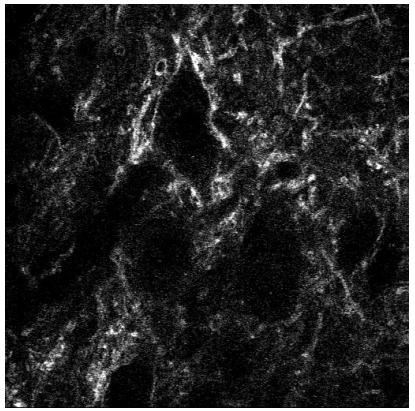
The selected rat locus-coeruleus recording follows native mu receptors labeled with the ligand-directed fluorescent probe. It uses ordinary rat tissue rather than requiring a genetically tagged receptor. Successive image planes reveal the distribution of the labeled structures through the living slice.

The receptor-binding guide directs the chemical reaction and then dissociates after the dye is attached. This leaves the binding site available for later receptor activation. The approach therefore differs from visualizing receptors only while a fluorescent antagonist remains bound and blocks them.

Functional tests verified that labeled cells retained opioid responses. Anatomical location could consequently be related to physiology in the same preparation. The movie establishes receptor visualization, while separate electrical measurements establish that the labeling procedure preserved the tested signaling function.



Arttamangkul et al. (2019), Fig. 3.



Native mu-receptor labeling in rat LC. Arttamangkul et al. (2019).

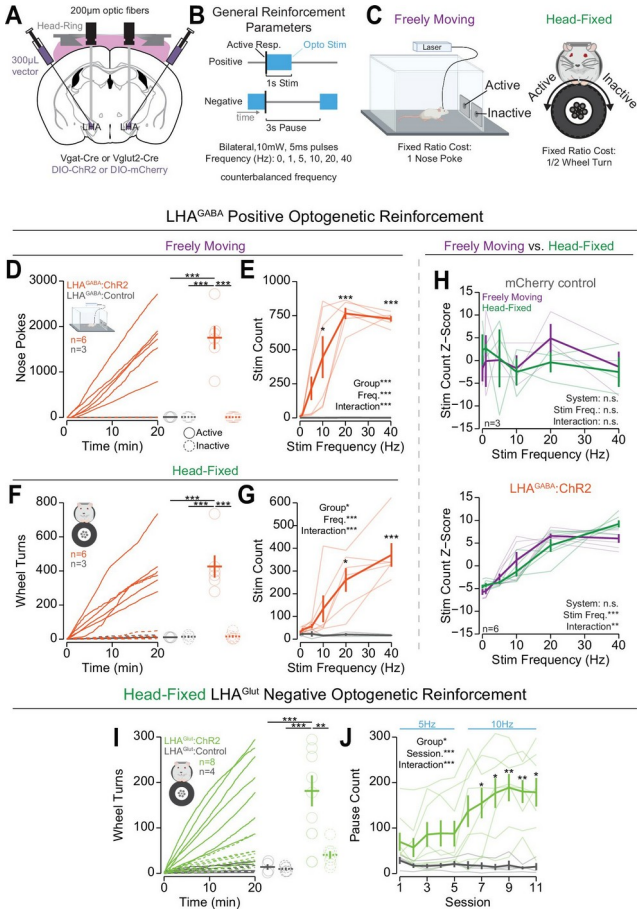
Play video

Behavioral treatment changes the consequences of use

Contingency management provides defined rewards for verified treatment goals, such as attendance or reduced substance use. The approach applies reinforcement to actions that support recovery. Its mechanism changes the available consequences of behavior rather than directly occupying a drug receptor.

Cognitive behavioral approaches identify cues, expectations, and situations that maintain use and develop alternative responses. Practical changes can also increase access to non-drug rewards and reduce exposure to triggering situations. The learned action is addressed within the person's daily environment.

For stimulant disorders, behavioral treatment remains especially important while medication and other biological approaches continue to be investigated. A candidate target in mice or a successful receptor experiment is not an established clinical medicine. Clinical outcomes require evidence from appropriately designed studies in people.



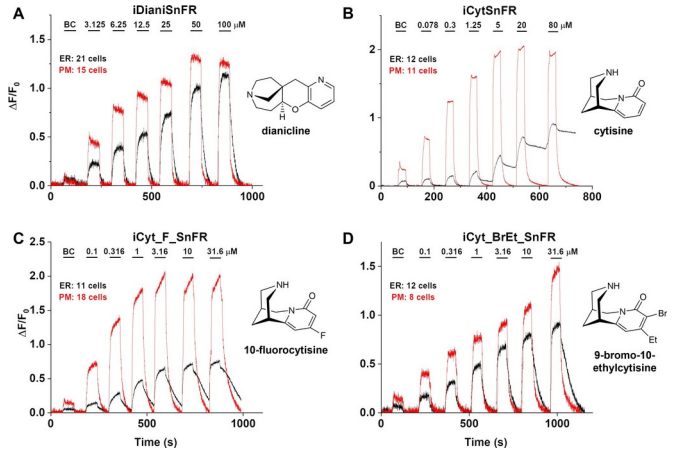
Gordon-Fennell et al. (2023), Fig. 3.

Dianicline reaches the endoplasmic reticulum

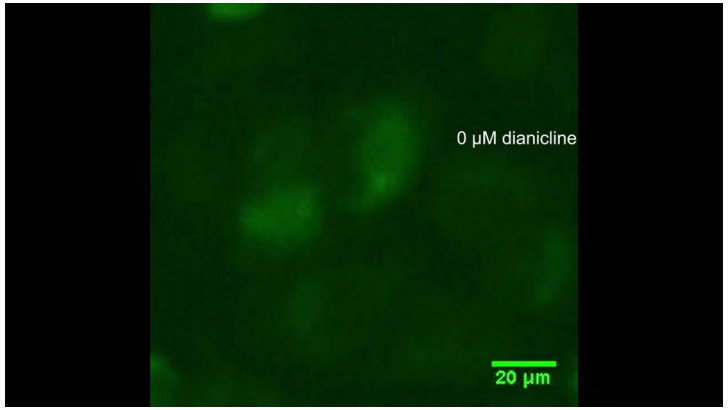
The biosensor researchers placed a drug sensor in the endoplasmic reticulum, an internal network involved in protein production and processing. They then applied dianicline outside the cell. A fluorescent response at this internal location required the compound to cross the plasma membrane and reach the sensor.

The selected original movie follows the endoplasmic-reticulum-targeted sensor in cultured cells. Brightness changes report local binding to the sensor, not a burst of neuronal firing. The preparation is a cellular drug-access assay rather than a recording from a person using a smoking-cessation medicine.

Comparing arrival and washout helped determine how rapidly the compound moved between compartments. Persistent internal exposure could differ from the concentration immediately outside the cell. The assay therefore measured a pharmacokinetic step that is hidden when only extracellular drug concentration is reported.



Nichols et al. (2022), Fig. 7.



Dianicline at an ER-targeted sensor. Nichols et al. (2022).

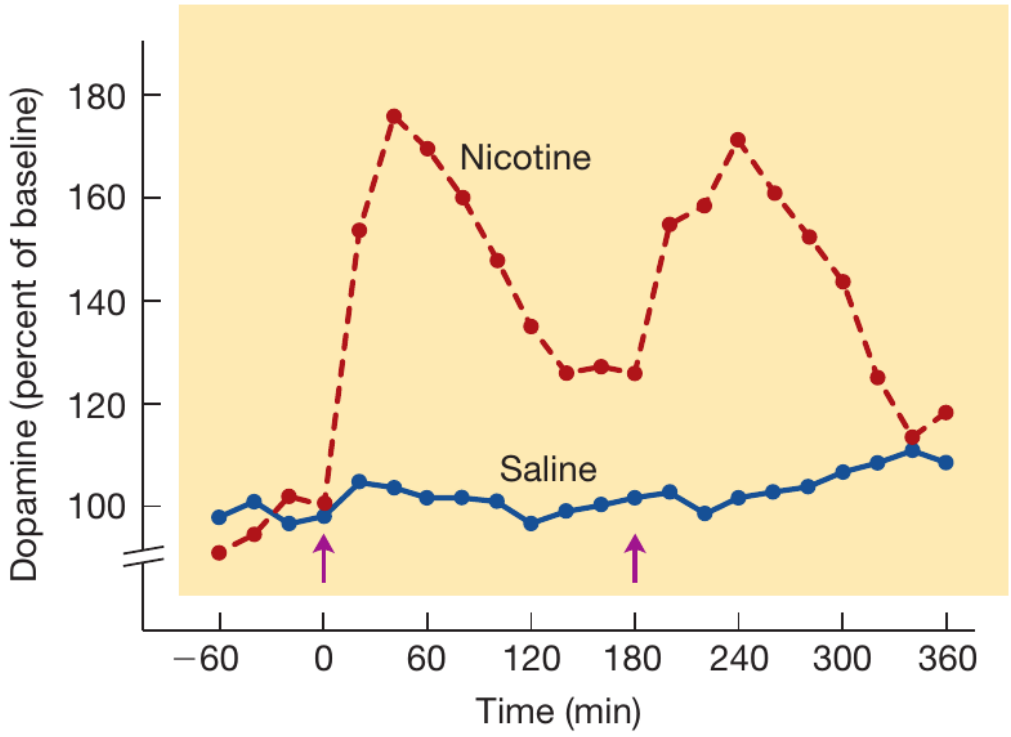
Play video

Nicotine replacement reduces withdrawal during cessation

Nicotine-replacement treatments provide nicotine through controlled formulations while avoiding tobacco combustion. Their delivery profiles differ from the rapid fluctuations associated with smoking. Reducing withdrawal can make it easier to practice abstinence and respond to cues without immediately returning to tobacco.

The treatment still activates nicotinic receptors, so its purpose is not instantaneous removal of all nicotine exposure. Instead, the form and time course are used to support cessation. Selection depends on clinical circumstances, preferences, and the pattern of symptoms being treated.

Behavioral support addresses routines, triggers, and learned expectations that medication does not directly erase. A cue can remain influential after withdrawal is reduced. Combining physiological stabilization with changes in daily actions addresses these distinct maintaining processes.



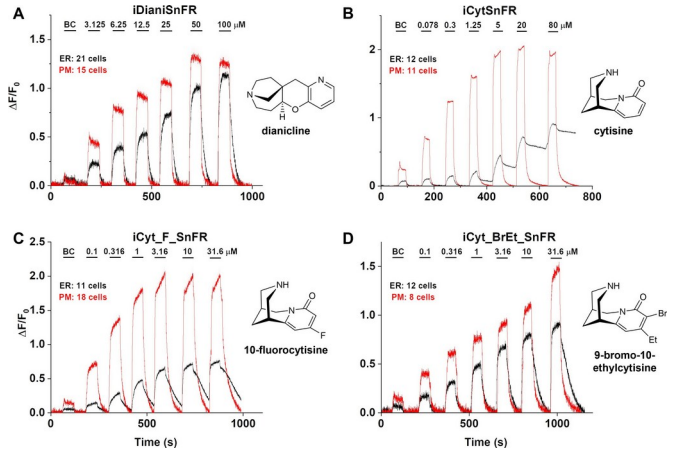
Carlson and Birkett (2022), Fig. 19.16, p.623.

A surface sensor separates entry from external arrival

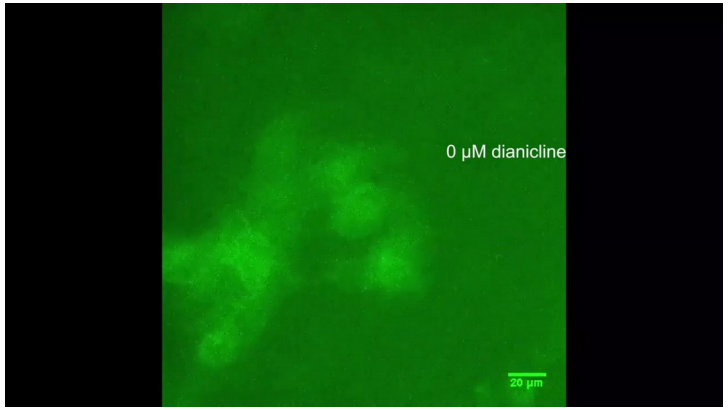
The paired dianicline recording follows a sensor at the plasma membrane rather than inside the endoplasmic reticulum. This sensor encounters the externally applied compound without requiring the same intracellular entry step. It provides a comparison for the delayed or persistent internal response.

Changes at the surface closely follow solution exchange, while internal measurements additionally depend on membrane crossing and compartmental access. The difference is interpreted using sensors with known binding properties. It is not assumed to arise from one natural receptor becoming more effective than another.

The two sensor locations therefore separate drug arrival outside the cell from access to internal binding sites. Their fluorescence is an exposure measure. Establishing therapeutic action still requires showing what the compound does to its biological targets and how that changes behavior or symptoms.



Nichols et al. (2022), Fig. 7.



Dianicline at a surface sensor. Nichols et al. (2022).

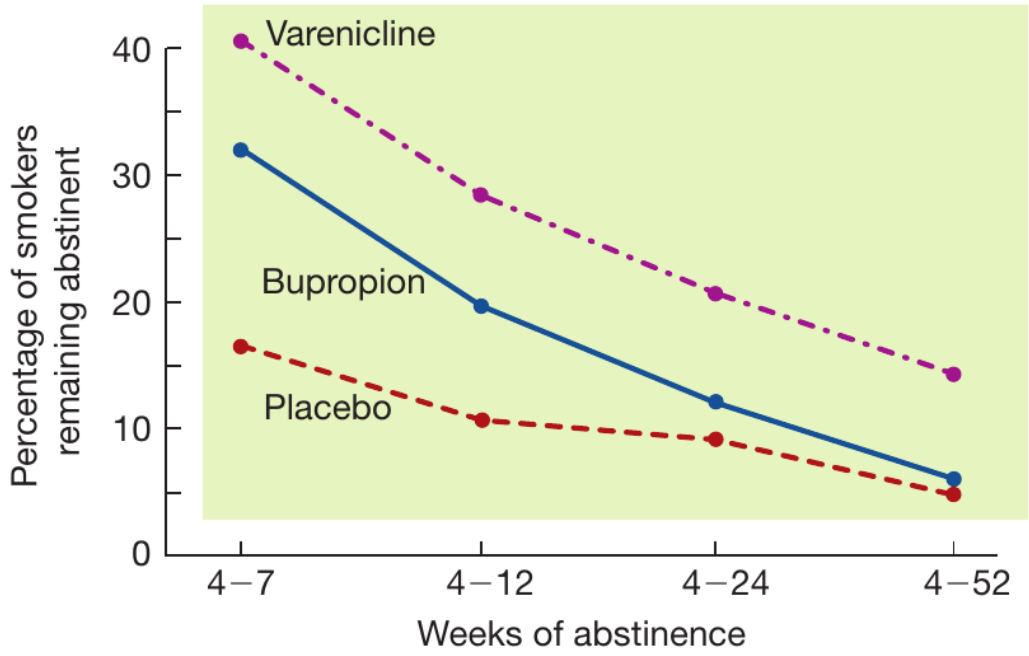
Play video

Cessation medicines act through distinct mechanisms

Varenicline partially activates selected nicotinic acetylcholine receptors and can reduce withdrawal while limiting the response to nicotine. Its partial agonism describes receptor activation relative to a fuller agonist in the relevant system. It does not mean the compound has no effect when nicotine is absent.

Bupropion affects monoamine signaling and also has actions relevant to nicotinic receptor function. Its full pharmacology differs from nicotine replacement and varenicline. Medicines that help the same clinical goal need not share one receptor-binding mechanism or the same adverse-effect profile.

Treatment selection also considers previous response, other conditions, and patient preference. A molecular description helps distinguish how a medicine acts, while clinical evidence establishes cessation outcomes. The choice cannot be inferred solely from which compound crosses a cellular membrane most quickly.



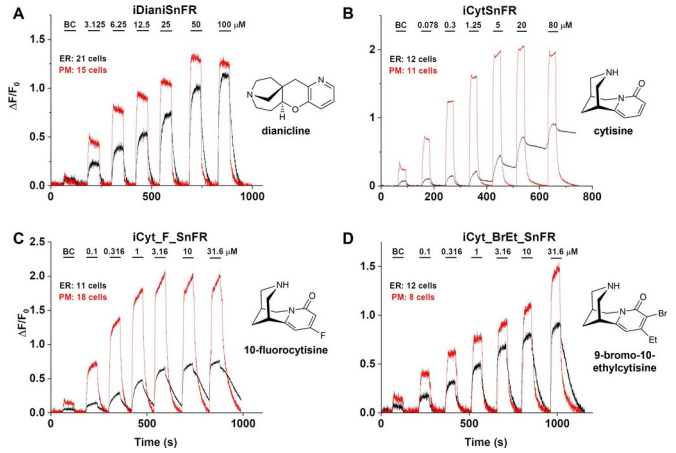
Carlson and Birkett (2022), Fig. 19.26, p.633.

Cytisine enters the internal compartment differently

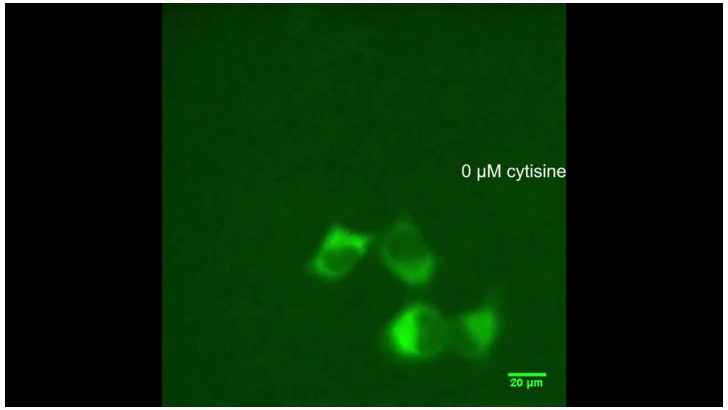
The original cytisine recording follows the endoplasmic-reticulum-targeted drug sensor. Like the dianicline experiment, the compound is supplied outside the cell and must reach the internal sensor. The brightness response therefore combines sensor binding with the time required for compartmental access.

The study found large differences in membrane permeation among the tested nicotinic partial agonists. Cytisine-related compounds did not all share the same entry and exit kinetics. Chemical similarity and receptor pharmacology therefore did not guarantee an identical intracellular exposure profile.

These differences can influence which receptor populations encounter a compound during a given exposure. Nicotinic receptors are assembled and processed within the secretory pathway before reaching the surface. Measuring internal drug access helps define the conditions under which such intracellular interactions are possible.



Nichols et al. (2022), Fig. 7.



Cytisine at an ER-targeted sensor. Nichols et al. (2022).

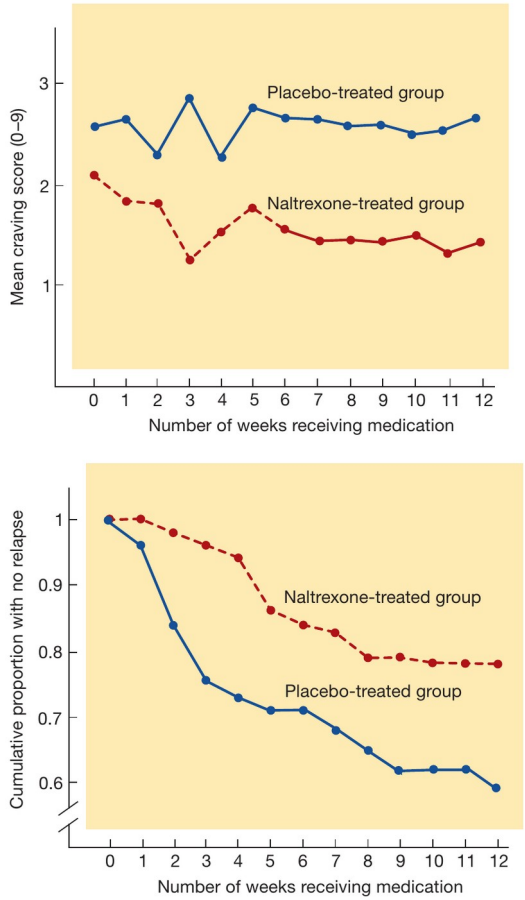
Play video

Alcohol medicines target different maintaining processes

Naltrexone blocks opioid receptors and can reduce alcohol-related reinforcement and heavy drinking in appropriate patients. Its action differs from making alcohol physically aversive through acetaldehyde accumulation. The relevant clinical effect is assessed through drinking outcomes and the person's treatment goals.

Acamprosate is used to support abstinence after withdrawal management. Its actions are associated with regulation of excitatory and inhibitory balance, but its complete therapeutic mechanism remains under study. Describing it as one established direct NMDA-receptor blockade would overstate what is known.

Disulfiram produces an aversive metabolic interaction, while behavioral approaches alter cues and consequences. These interventions target different parts of the maintaining cycle. Medical stabilization, reduced reinforcement, and support for sustained abstinence should therefore be distinguished rather than treated as one generic anti-alcohol effect.



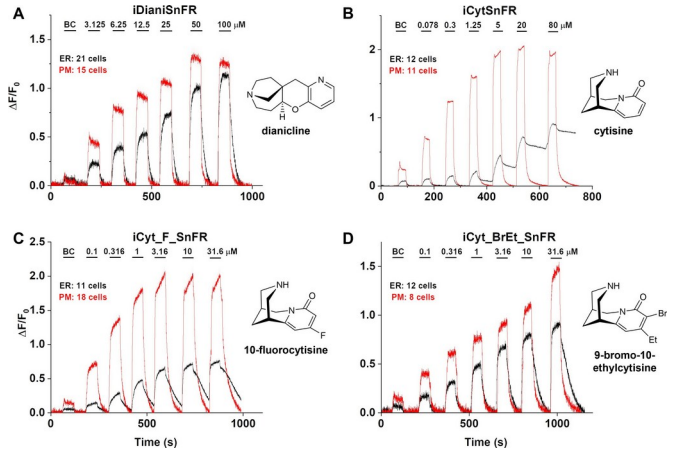
Carlson and Birkett (2022), Fig. 19.27, p.634.

A surface cytisine signal supplies the comparison

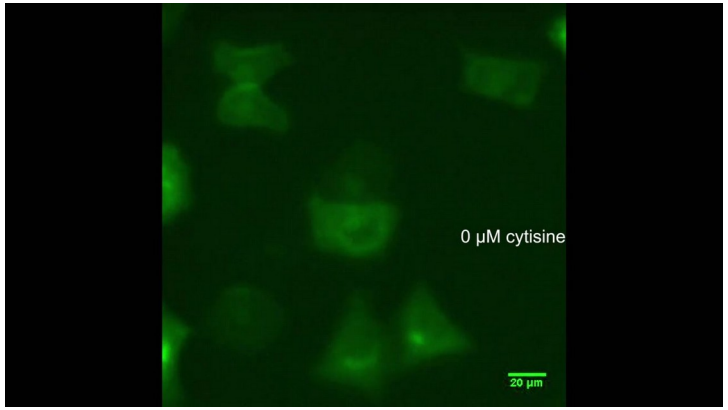
The paired cytisine movie follows the plasma-membrane-targeted sensor. Its response records external arrival at the cell boundary. Comparing it with the internal sensor helps identify whether a slower intracellular signal reflects entry through a membrane rather than slow delivery to the culture.

The study also measured sensor binding properties so that drug recognition could be separated from permeability. A dim or delayed signal alone could otherwise reflect either low access or weak sensor binding. The comparison uses these measurements to interpret the compartment-specific time courses.

Drug distribution is therefore a measurable part of pharmacology rather than an assumed preliminary step. Two compounds can act at related receptors yet expose internal and surface populations differently. The demonstrated result concerns access in the tested cells, separate from clinical cessation effectiveness.



Nichols et al. (2022), Fig. 7.



Cytisine at a surface sensor. Nichols et al. (2022).

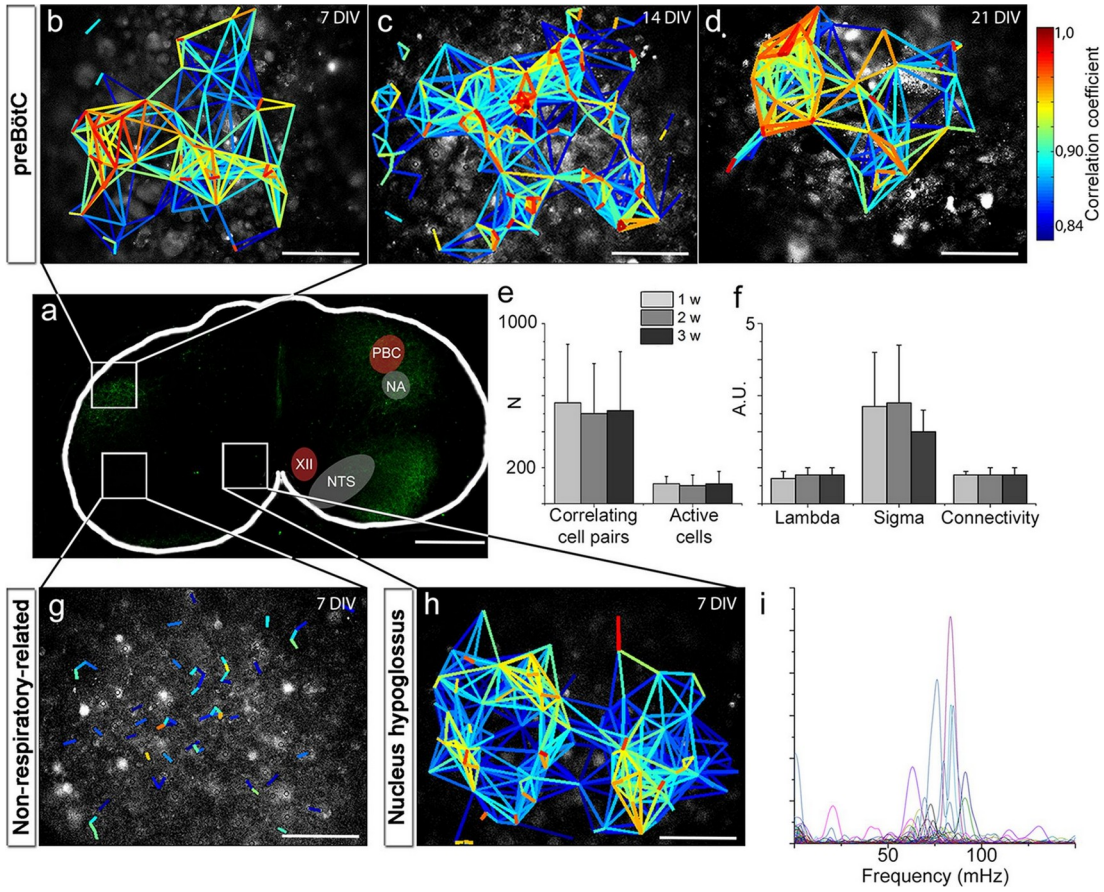
Play video

Recovery includes acute rescue and long-term prevention

An opioid overdose can suppress breathing and requires urgent emergency action, including naloxone where available. Reversing receptor activation addresses the immediate physiological danger. It does not by itself treat the longer-term learned and biological processes that sustain opioid use disorder.

Maintenance medication, behavioral treatment, and practical support address recurrence and daily functioning. Brain-stimulation approaches are also being investigated for selected substance-related conditions, but circuit plausibility alone does not establish routine clinical effectiveness. Their outcomes must be evaluated separately from acute rescue.

A return to use after reduced tolerance can carry serious risk because the earlier exposure no longer produces the same physiological response. Continued care therefore includes prevention and follow-up rather than treating one period of abstinence as permanent removal of vulnerability. Recovery can require sustained support across changing circumstances.



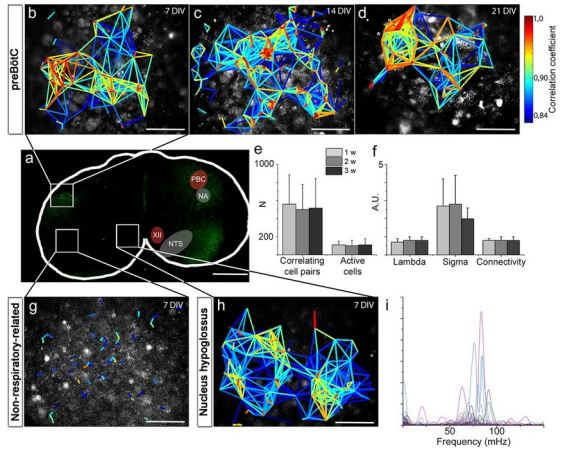
Forsberg et al. (2016), Fig. 4.

Opioid activation can suppress a respiratory network

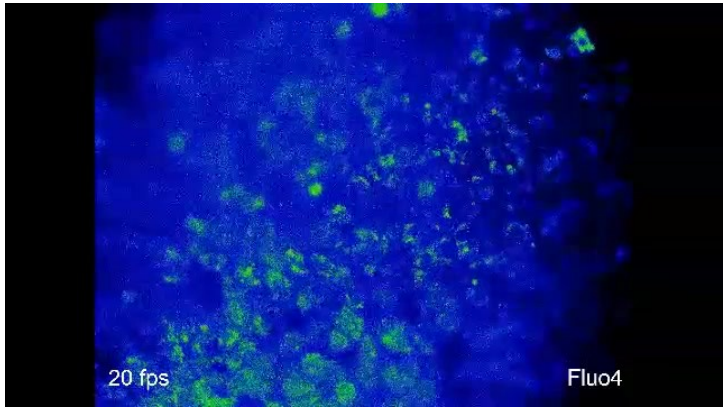
Researchers maintained neonatal rodent brainstem tissue containing the pre-Bötzinger complex, a network involved in generating respiratory rhythm. Calcium imaging followed its recurring activity in culture. This preparation allowed the direct effect of an opioid-receptor agonist to be observed without whole-animal behavioral changes.

The selected original recording shows reduced network activity after DAMGO, a mu-receptor agonist, is added. The fluorescence follows cellular calcium events rather than chest movement. Suppression of this rhythm-related preparation provides a cellular example of why opioid action can threaten respiratory function.

The experiment does not show naloxone rescue or maintenance treatment. Its measured event is inhibition of a defined neonatal respiratory network by receptor activation. Emergency antagonism and long-term treatment address different consequences of opioid exposure than the initial network effect recorded here.



Forsberg et al. (2016), Fig. 4.



Mu agonist suppresses brainstem-network activity. Forsberg et al. (2016).

Play video

Key takeaways

Reinforcement depends on the consequence. Obtaining reward and removing withdrawal can both strengthen use; punishment, negative reinforcement, tolerance, dependence, and impaired control are different concepts.

Drug mechanisms differ before they converge. Opioids, stimulants, nicotine, alcohol, and cannabinoids begin at different receptors, transporters, or channels even when several influence dopamine-related learning.

Cues retain learned value. Places, odors, internal sensations, and stress can renew responding after the drug is absent; intoxication, withdrawal, and cue-triggered seeking require separate measurements.

Receptor location changes exposure and signaling. Membrane entry, phosphorylation, endocytosis, recycling, and intracellular compartments determine which targets a drug reaches and how their responses change over time.

A task result requires its actual preparation. Calcium fluorescence, movement, receptor labeling, self-administration, and clinical recovery measure different events; a cellular or animal finding is not by itself a clinical treatment outcome.

Recovery addresses several maintaining processes. Maintenance medicines stabilize opioid exposure, rescue antagonists reverse acute toxicity, cessation and alcohol medicines target distinct mechanisms, and behavioral care changes cues, consequences, and daily functioning.

Carlson & Birkett (2022); primary studies in notes